CompOmics: Proteomics Informatics from A(I) to Z(witterion)



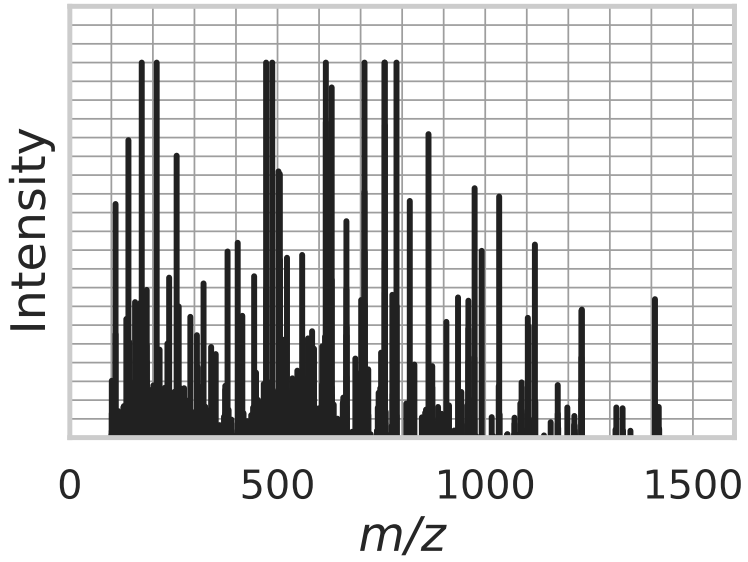
VIB-UGENT CENTER FOR MEDICAL BIOTECHNOLOGY

GHENT UNIVERSITY

CompOmics

15/04/2026

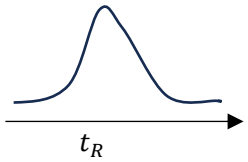
From biological research questions to data, identification + quantification, and biological insights

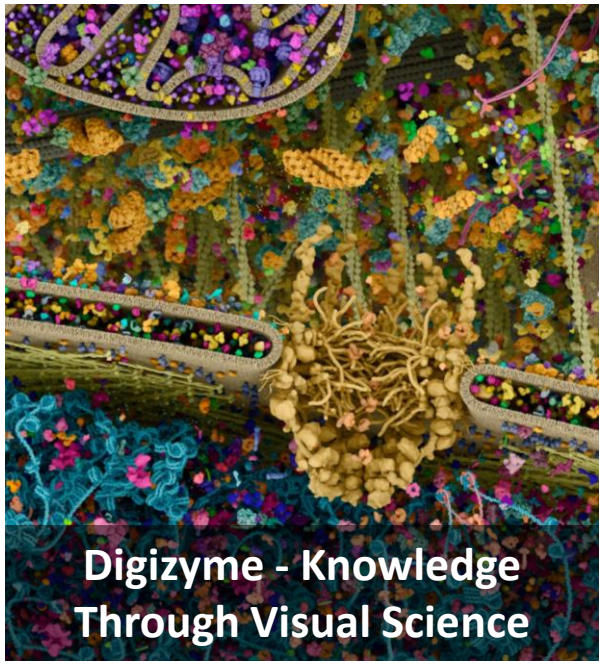


Identification

+

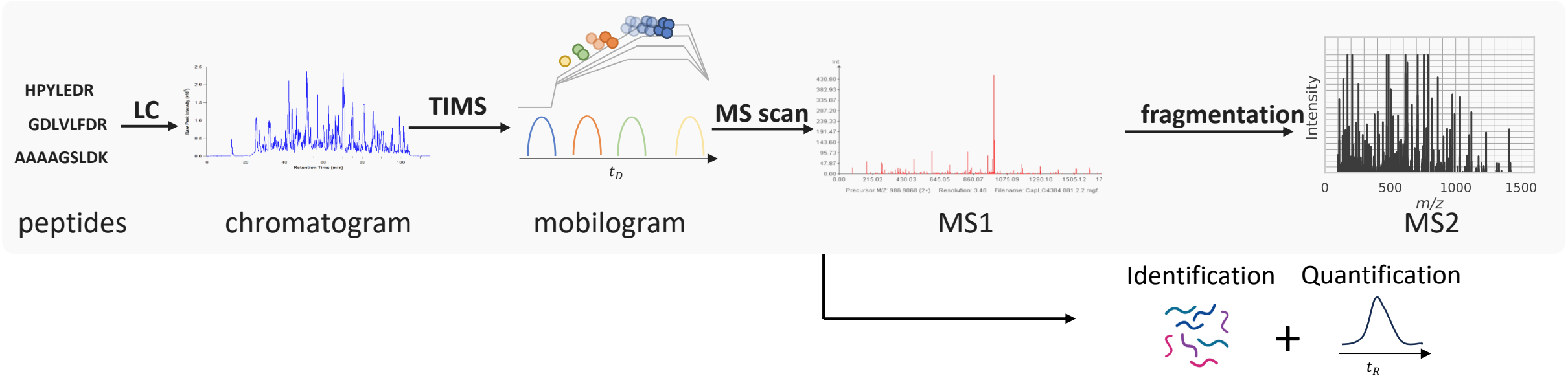
Quantification





MS-based proteomics from data to identification & quantification

experimental

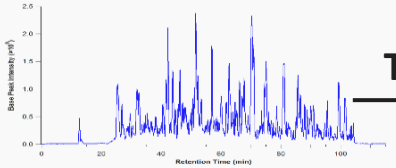


MS-based proteomics from data to identification & quantification

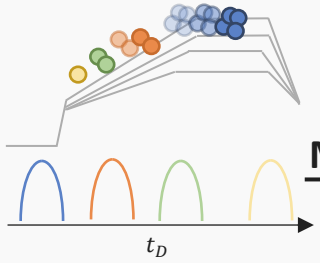
experimental

HPYLEDR
GDLVLFDR
AAAAGSLDK

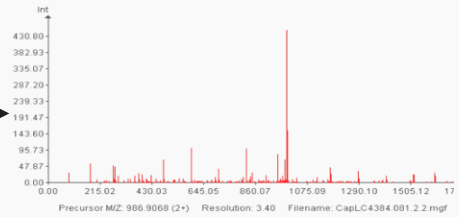
LC



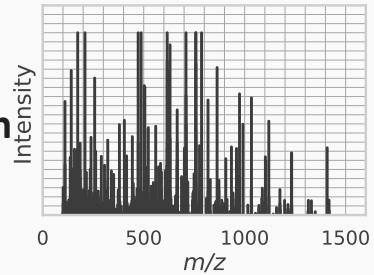
TIMS



MS scan



fragmentation

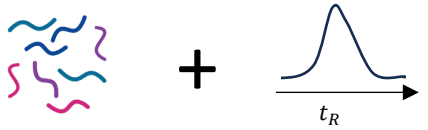


peptides | chromatogram | mobilogram | MS1 | MS2

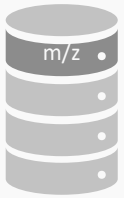
$$HS = \left(\sum_{i=0}^{n} I_i P_i\right) N_b!\, N_y!$$

Scoring function

Identification + Quantification

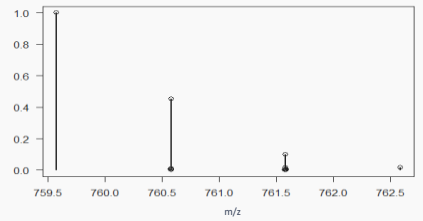


in silico



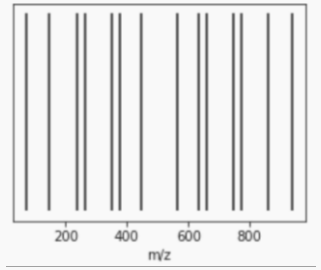
calculate

mass



in silico

MS/MS



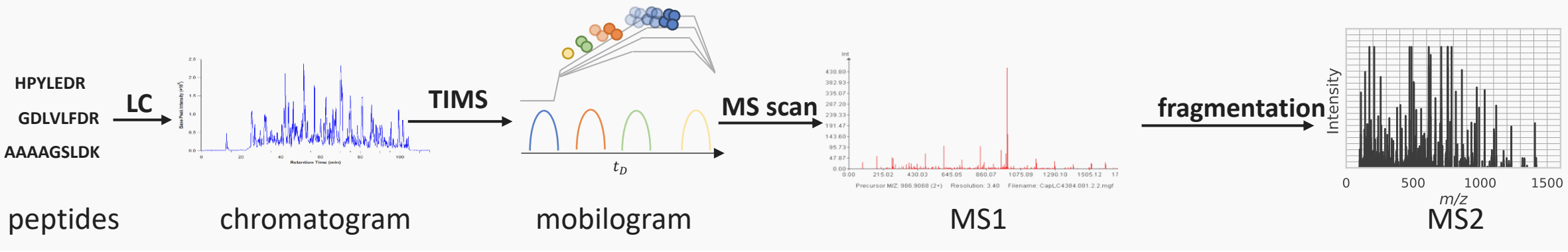
fasta database

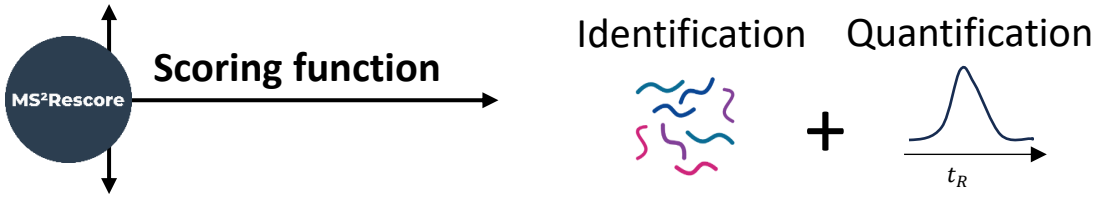
precursor mass (and isotopic distribution)

theoretical spectra

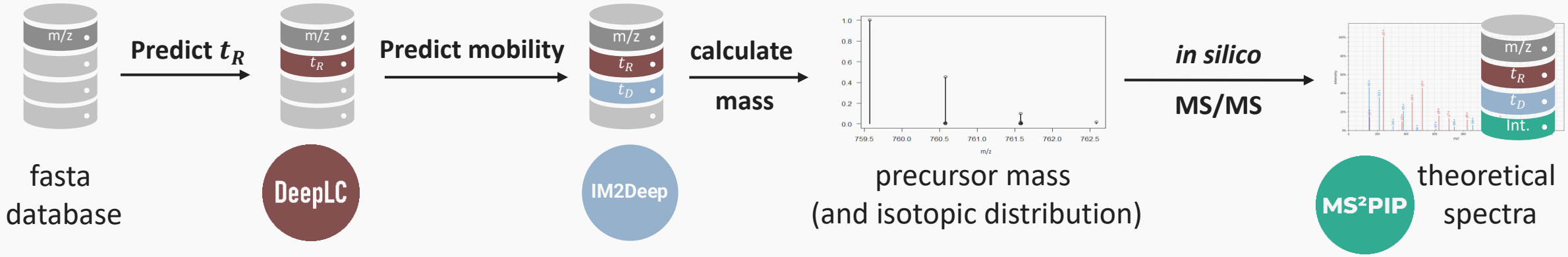
Predicted behavior for better *in silico* representation of the signal

experimental



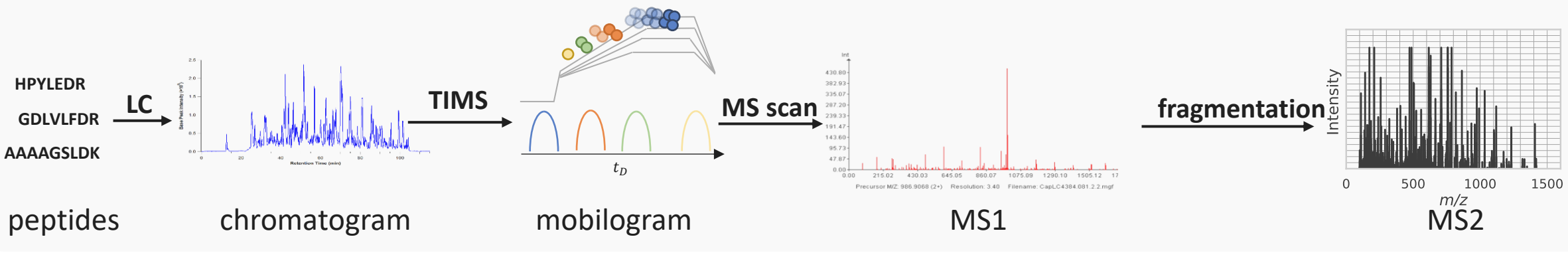


in silico



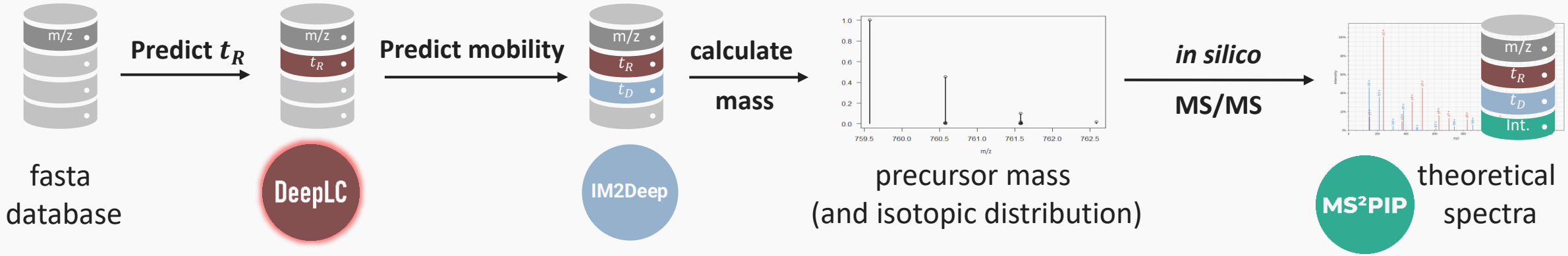
Predicted behavior for better *in silico* representation of the signal

experimental

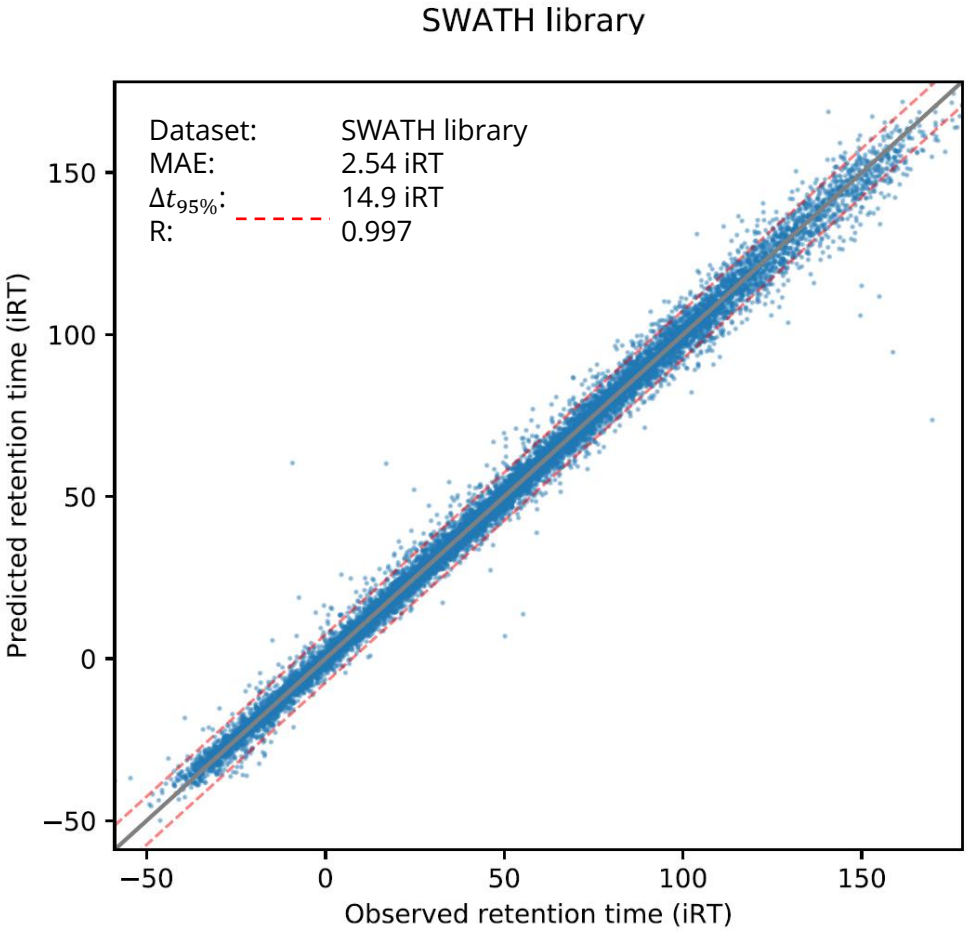


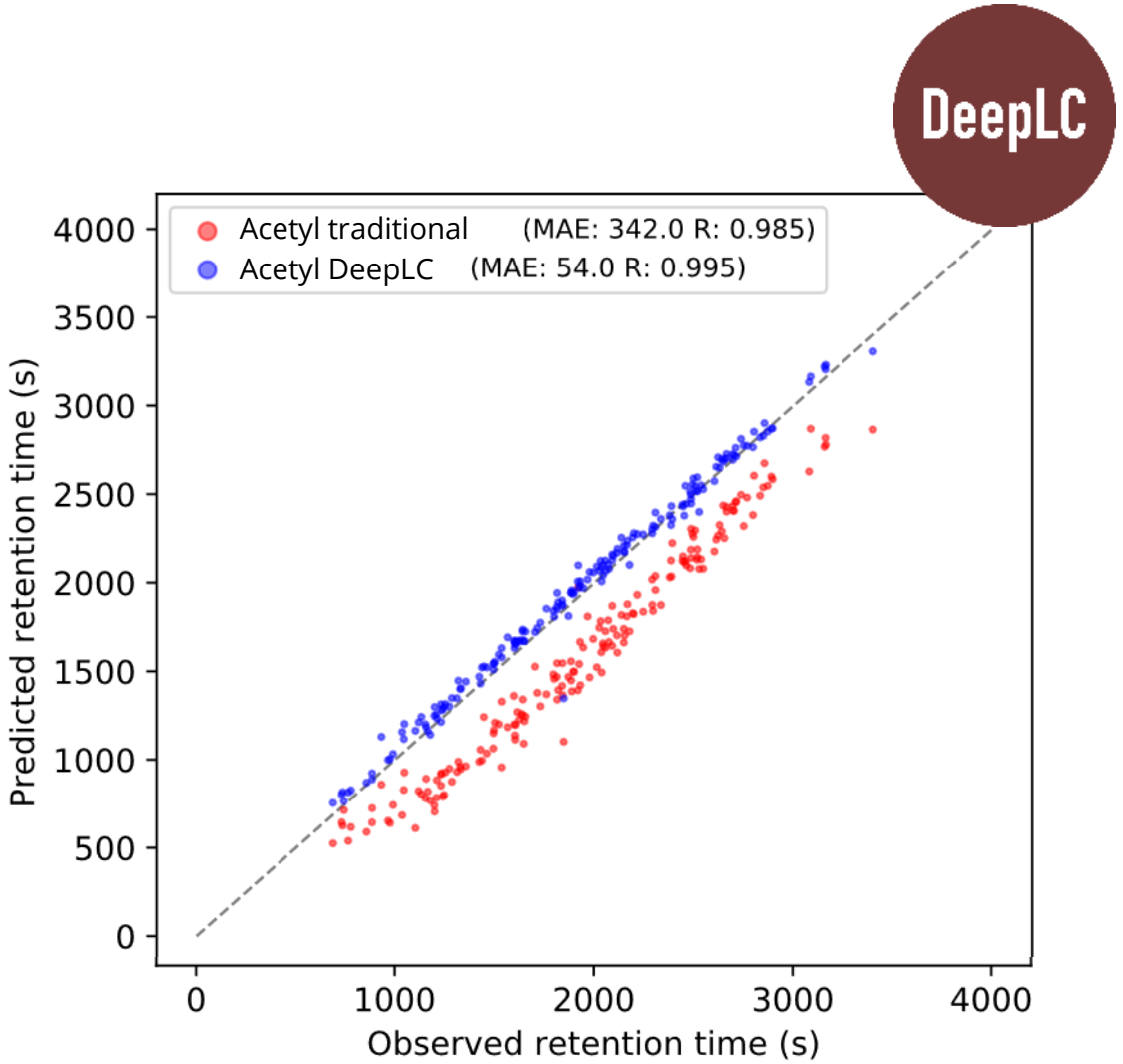
MS²Rescore — **Scoring function** → Identification + Quantification (t_R)

in silico



DeepLC can accurately predict retention times of (modified) peptides; more than just a retention time shift

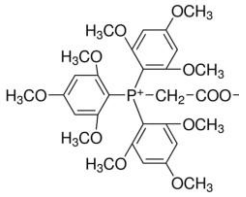




Bouwmeester *et al.* Nature Methods (2021)

Transfer learning for TMPP labels shows the potential for modifications very different from previous training data

(tris(trimethoxyphenyl)phosphine)-labels

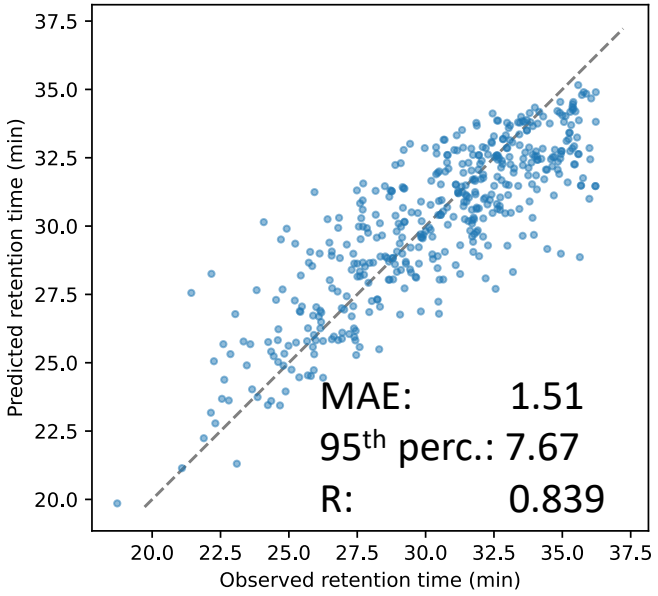


Université de Strasbourg

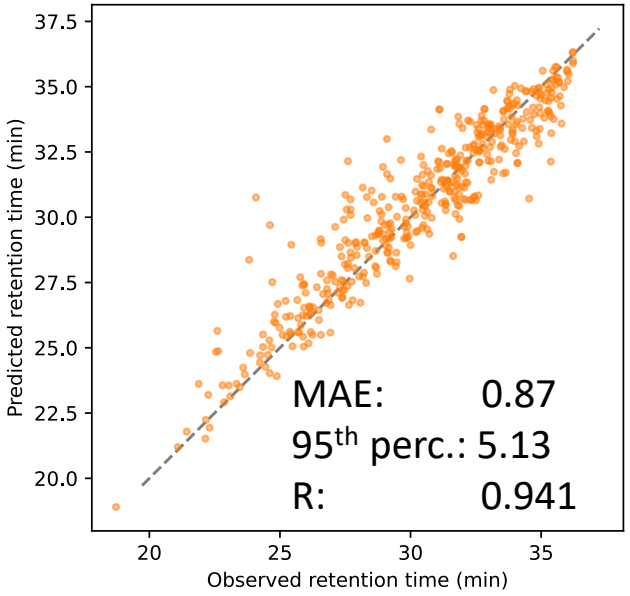
Christine Carapito
Magali Rompais



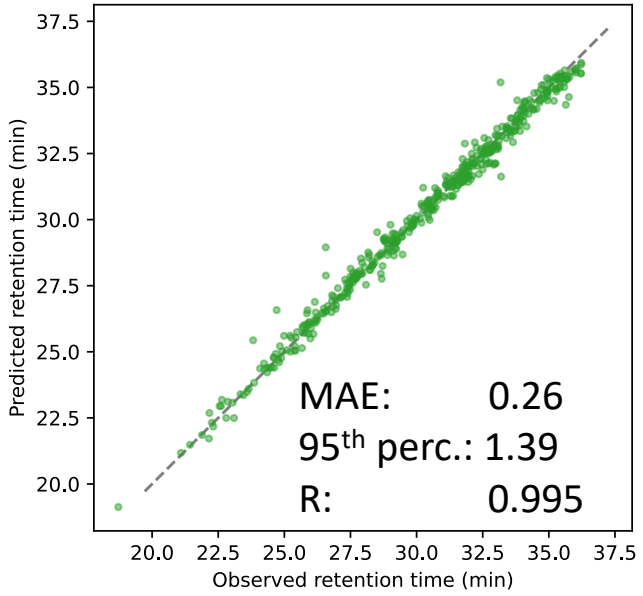
Train new model



Calibration

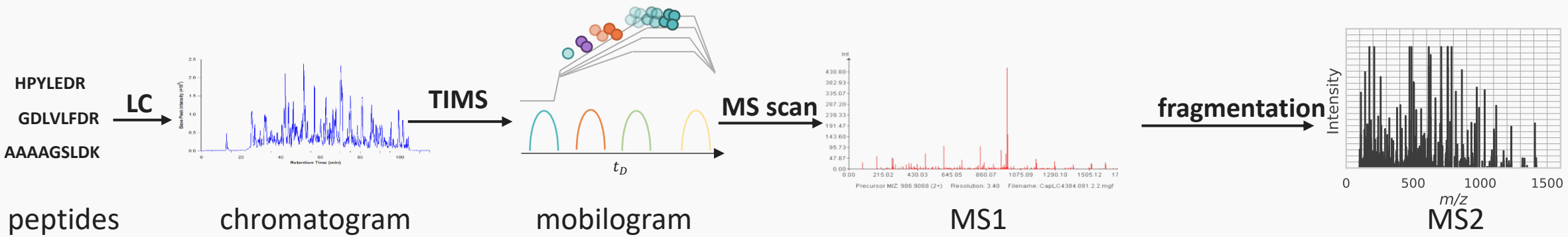


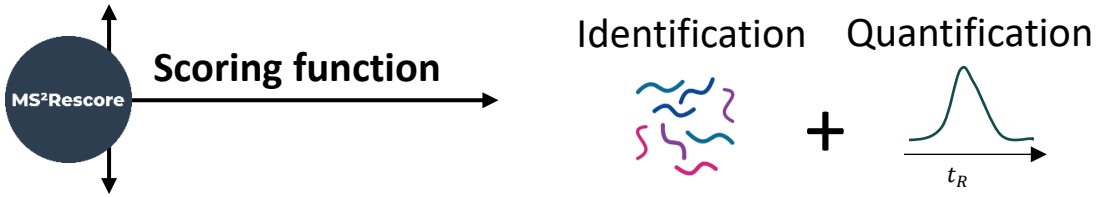
Transfer learning



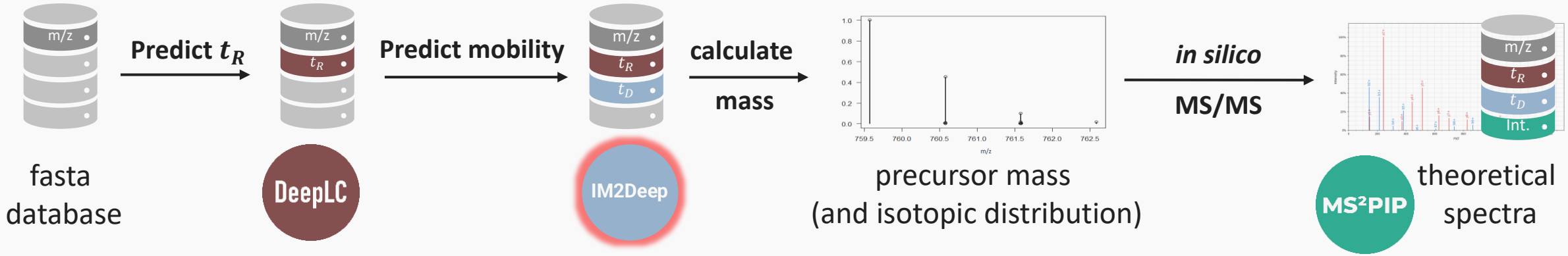
Predicted behavior for better *in silico* representation of the signal

experimental



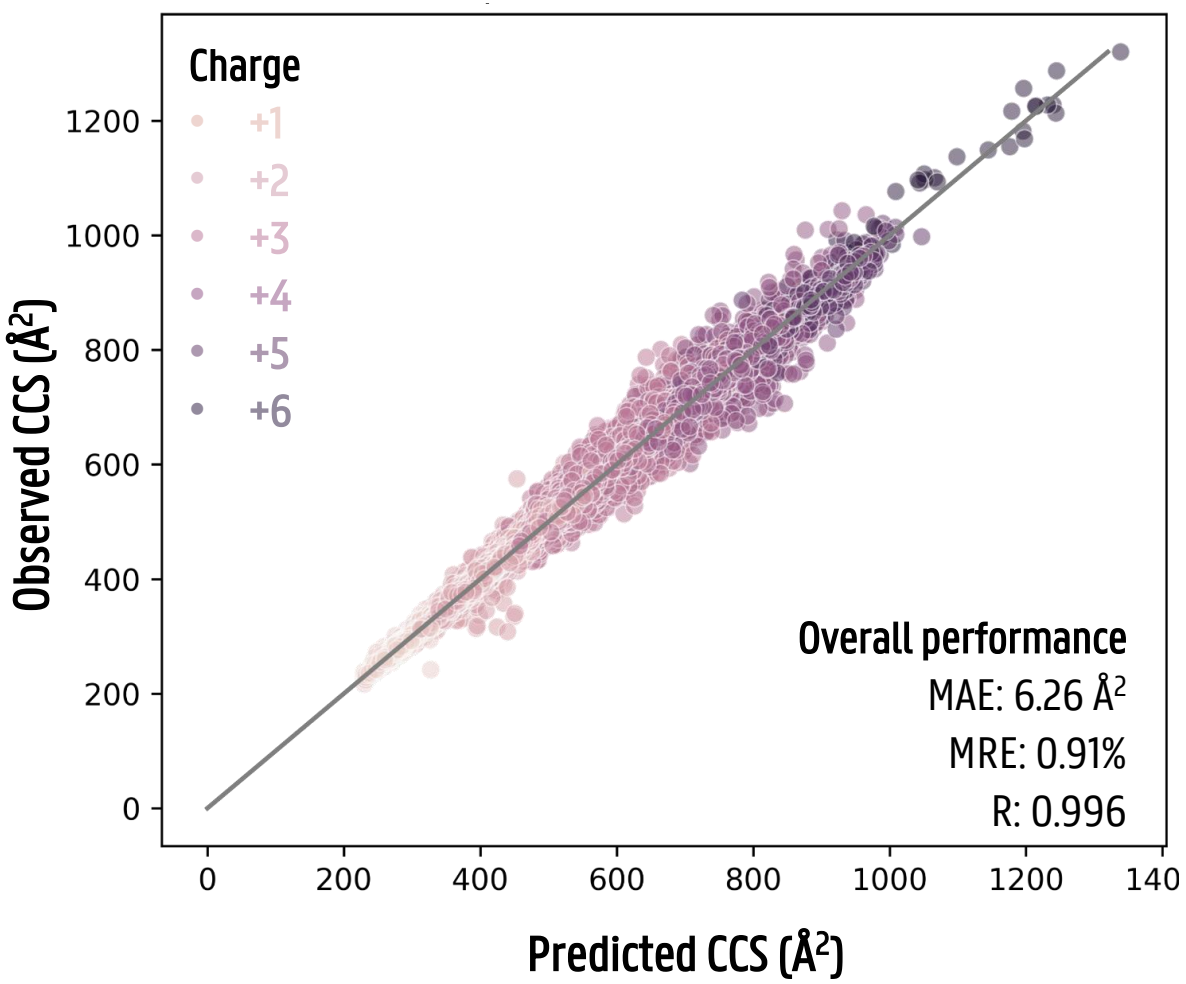


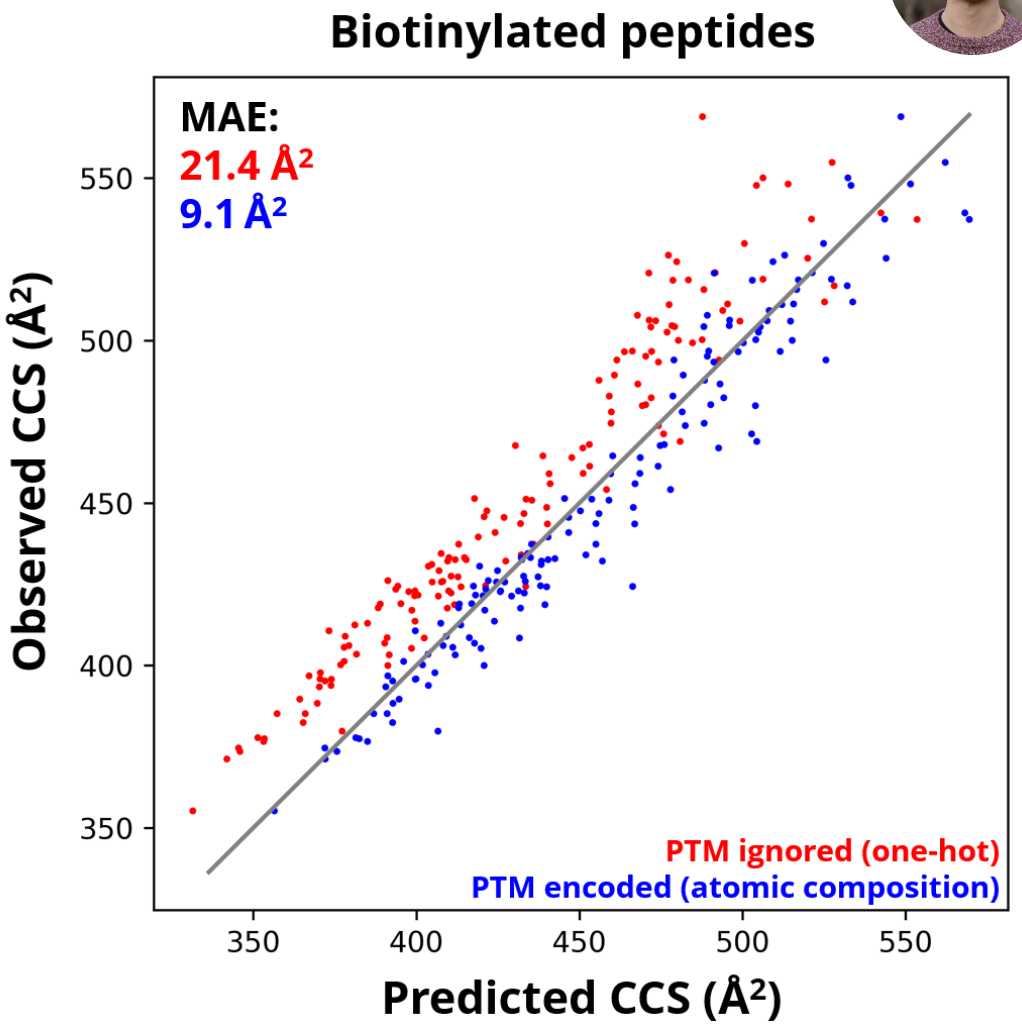
in silico



Encoding modified peptides by their atomic composition allows accurate CCS predictions for PTMs



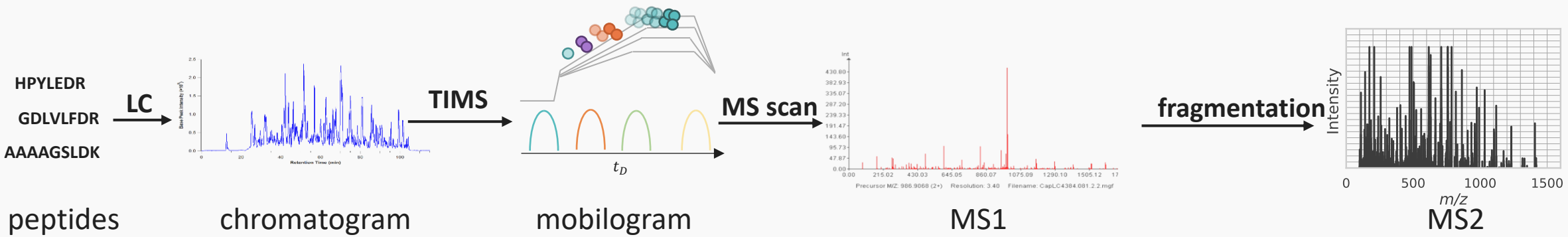




Declercq*, Devreese* *et al.* (2024) bioRXiv. doi:10.1101/2024.05.29.596400

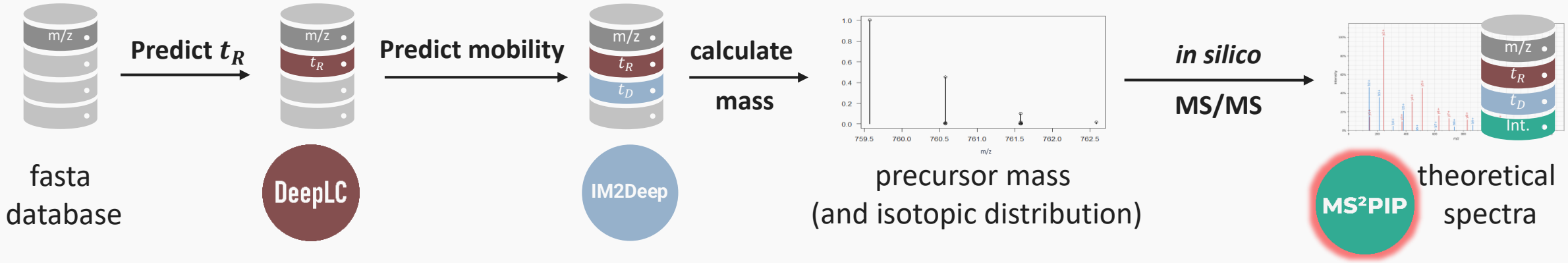
Predicted behavior for better *in silico* representation of the signal

experimental

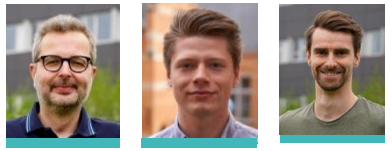


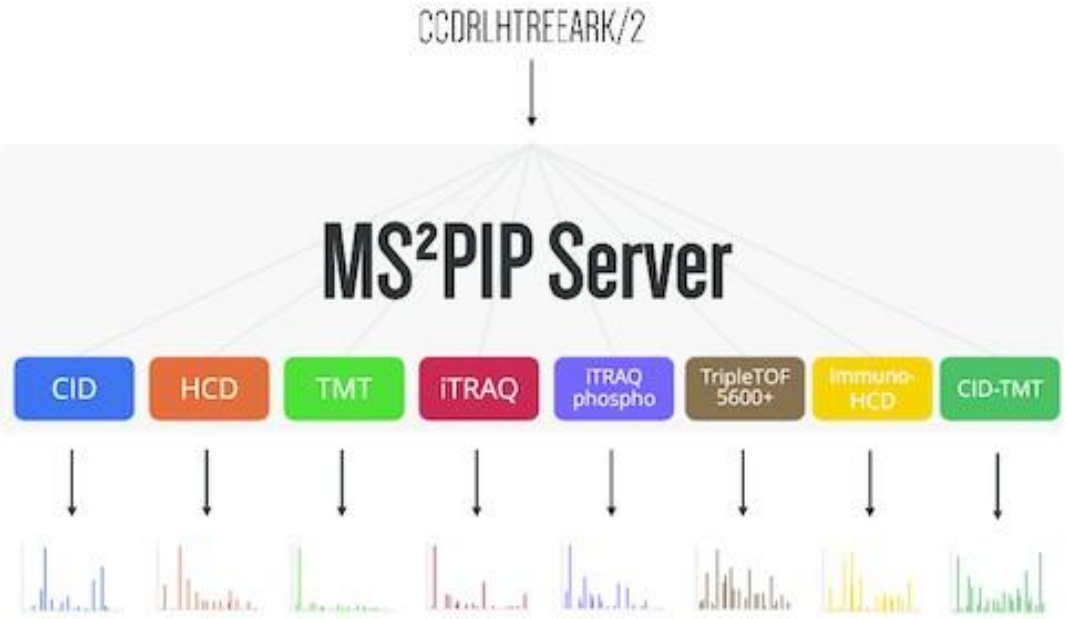
MS²Rescore — **Scoring function** → Identification + Quantification (t_R)

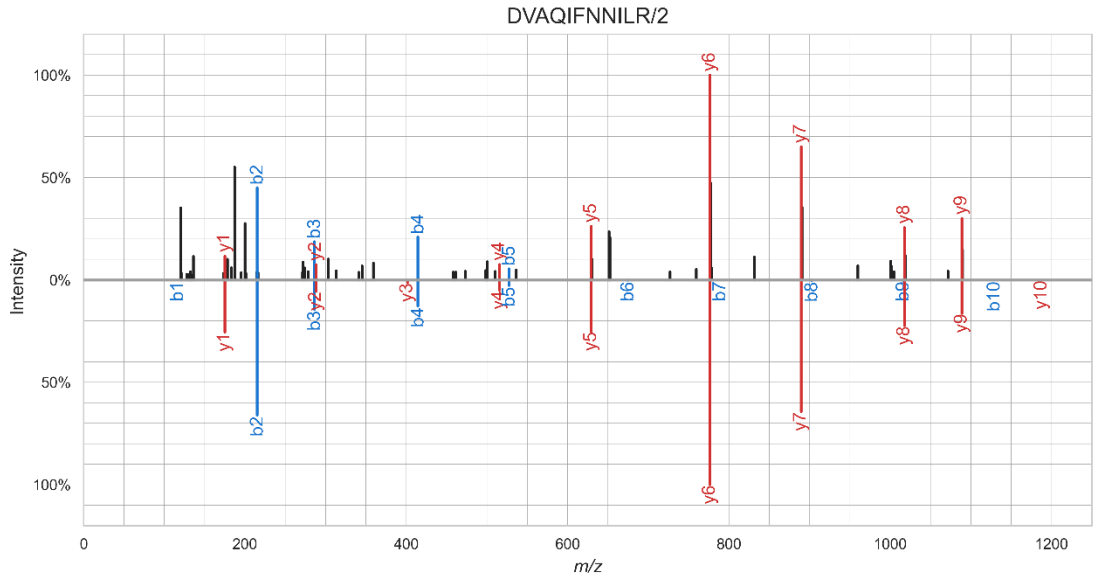
in silico



Our MS²PIP fragmentation models accurately predict peptide fragmentation behavior in varying conditions



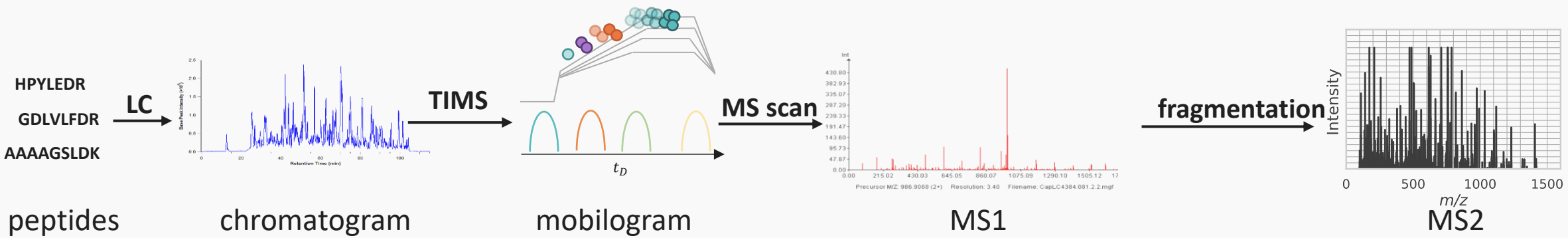


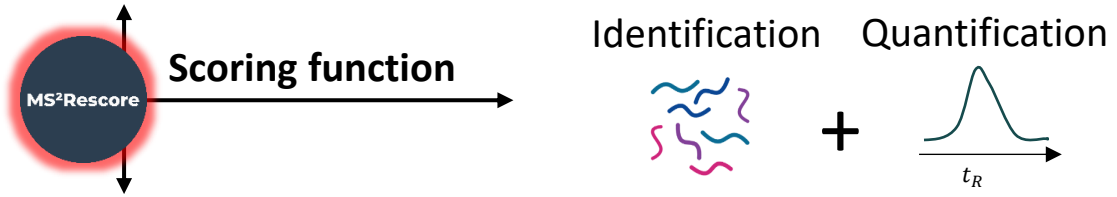


Degroeve, Bioinformatics (2013)
Degroeve, Nucleic Acids Research (2015)
Gabriels, Nucleic Acids Research (2019)
Declercq, Nucleic Acids Research (2023

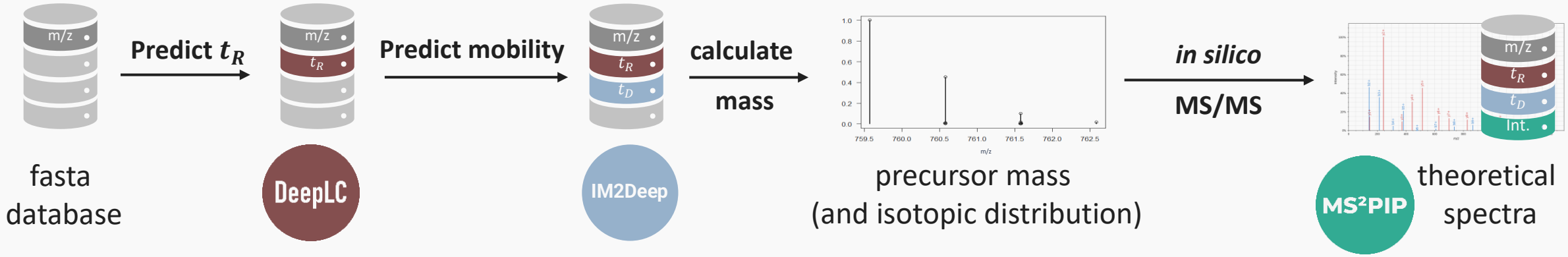
Predicted behavior for better *in silico* representation of the signal

experimental



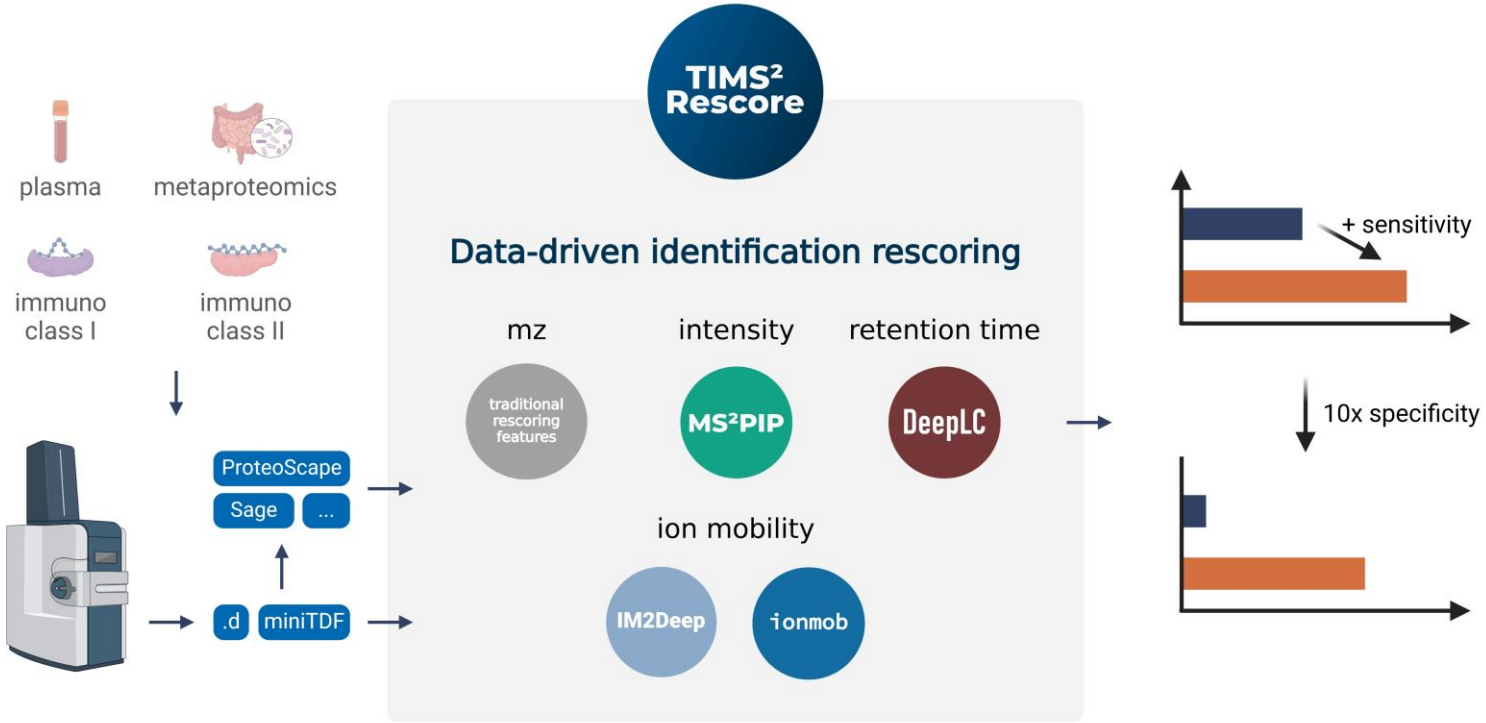


in silico



TIMS²Rescore combines the full workflow in ProteoScape (and is also available as a Python package)

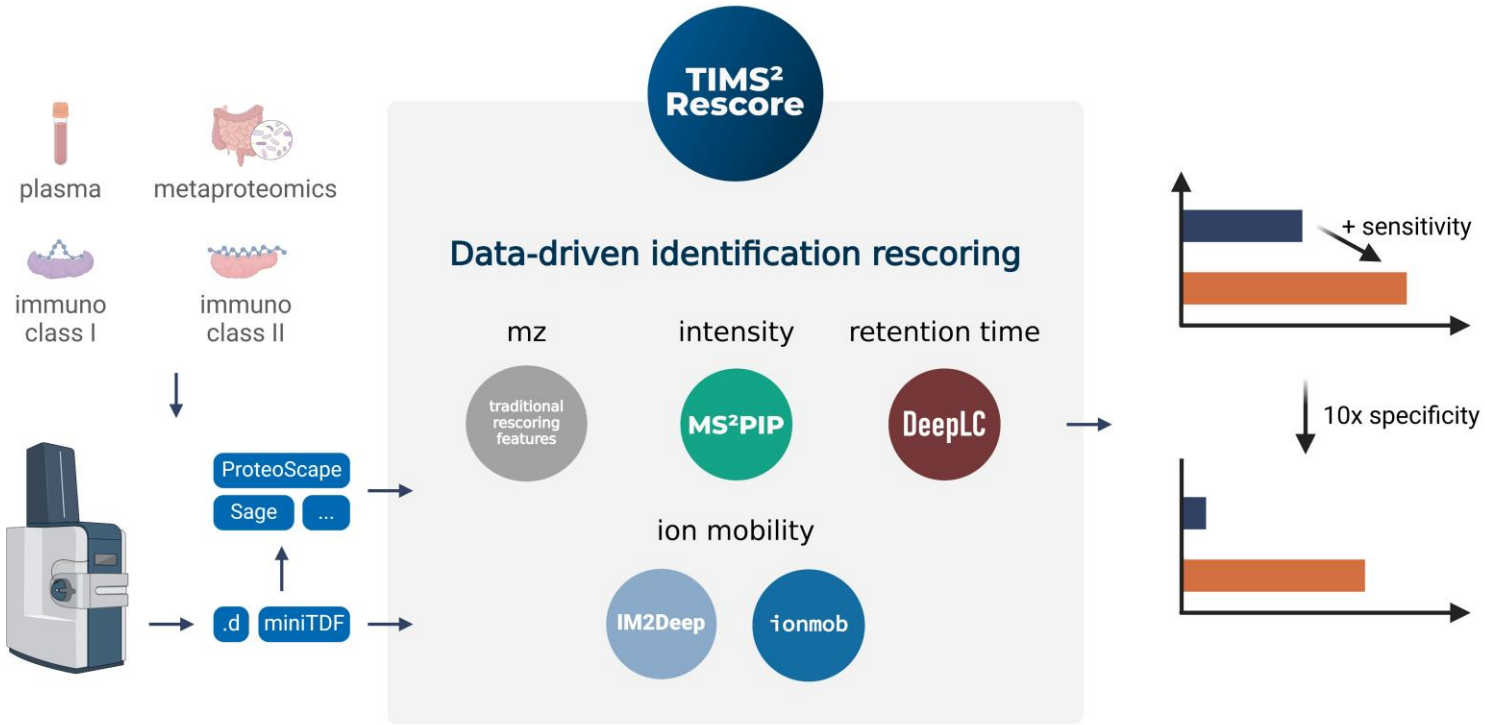


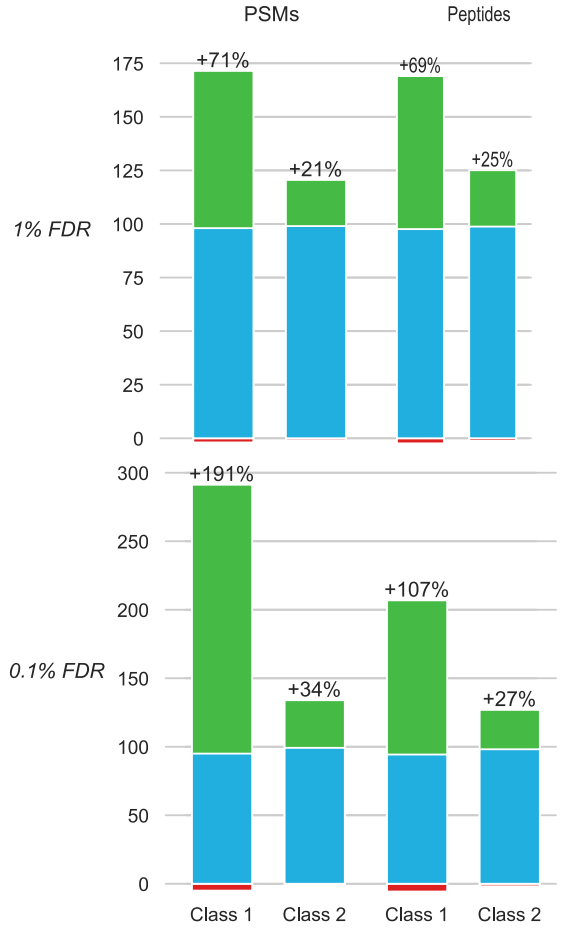


Arthur Declercq*, Robbe Devreese*, [...] Robbin Bouwmeester, Lennart Martens, Ralf Gabriels (2024) bioRXiv. doi:10.1101/2024.05.29.596400

TIMS²Rescore combines the full workflow in ProteoScape (and is also available as a Python package)



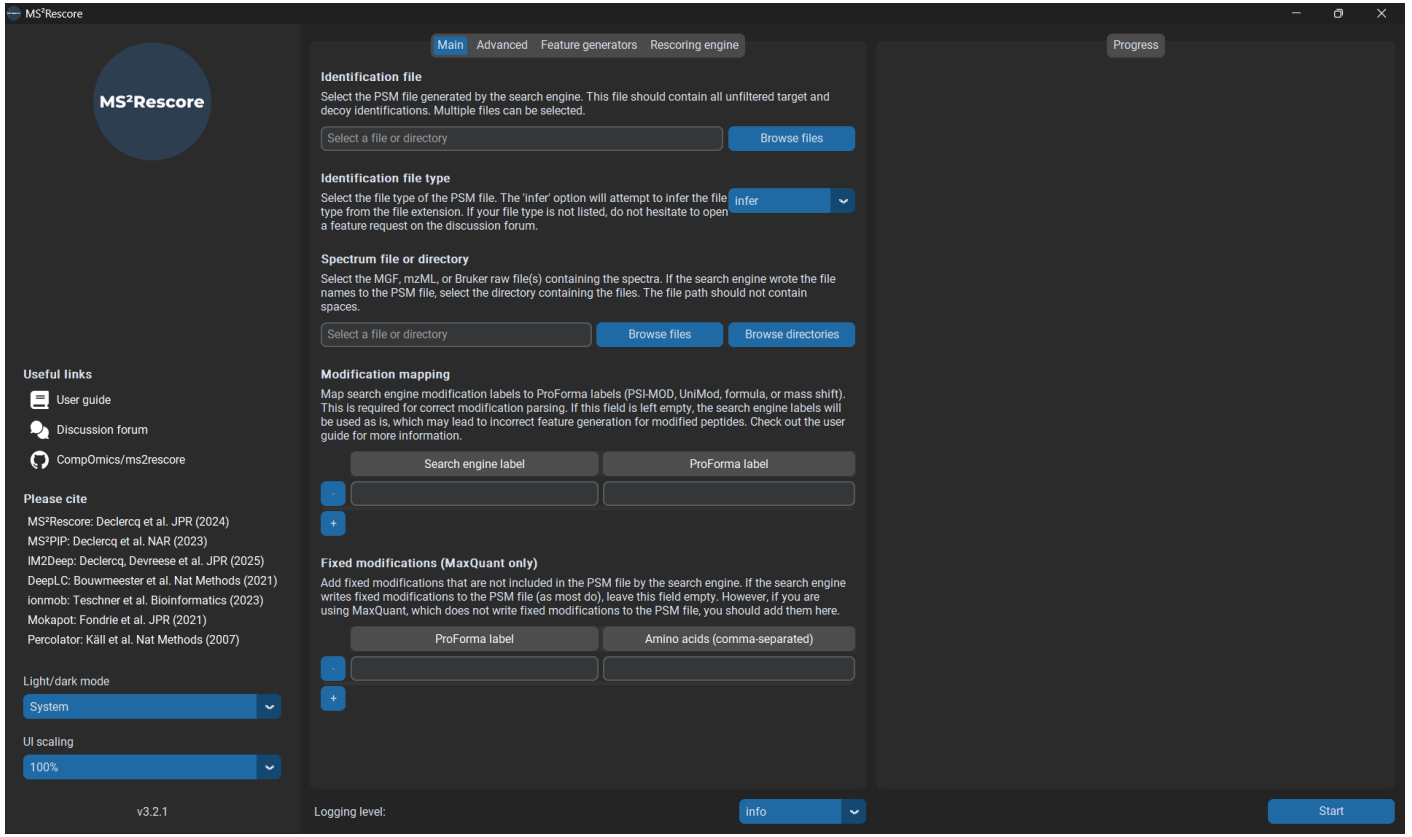




Arthur Declercq*, Robbe Devreese*, [...] Robbin Bouwmeester, Lennart Martens, Ralf Gabriels (2024) bioRXiv. doi:10.1101/2024.05.29.596400

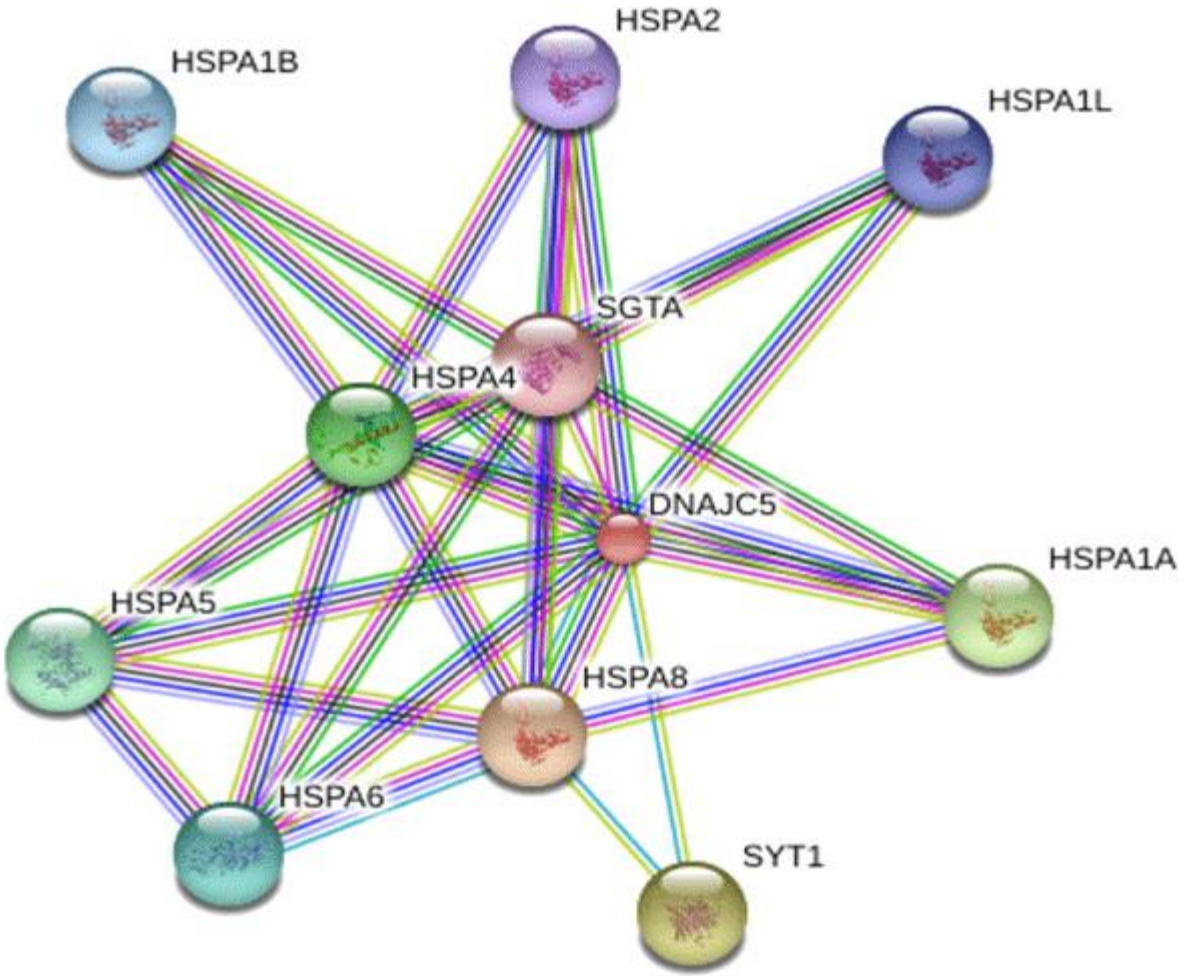
MS²Rescore is an open and freely available (and is also available as a Python package)





https://github.com/compomics/ms2rescore

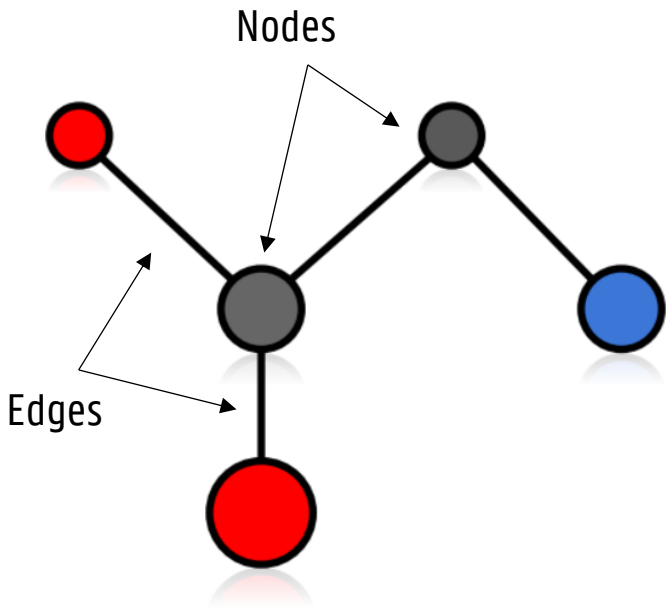
Networks or graphs to represent peptides; the next big thing in behavior prediction

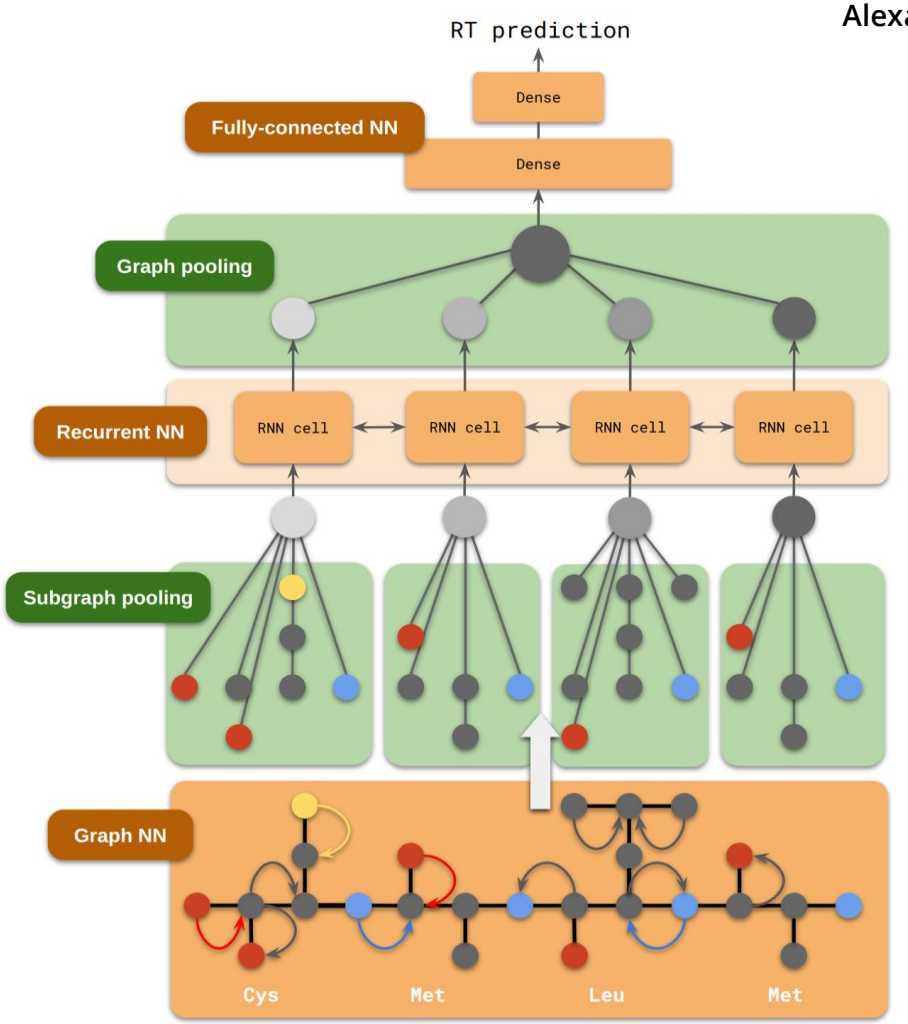


An improved representation with graph neural networks; respecting full structural impact



Alexander Kensert





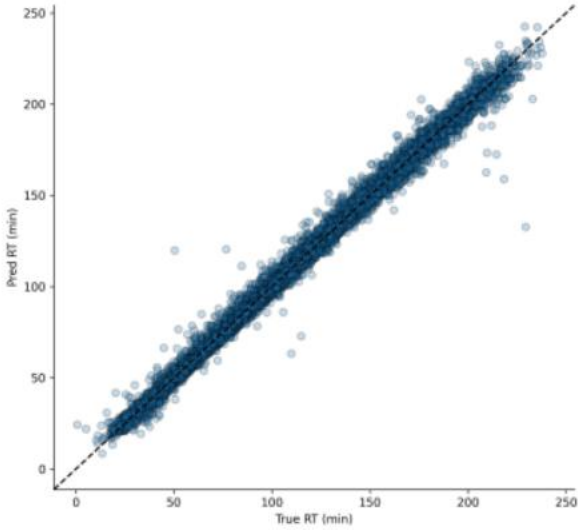
Graph representation slightly improves performance; more importantly structure contribution interpretation



Alexander Kensert

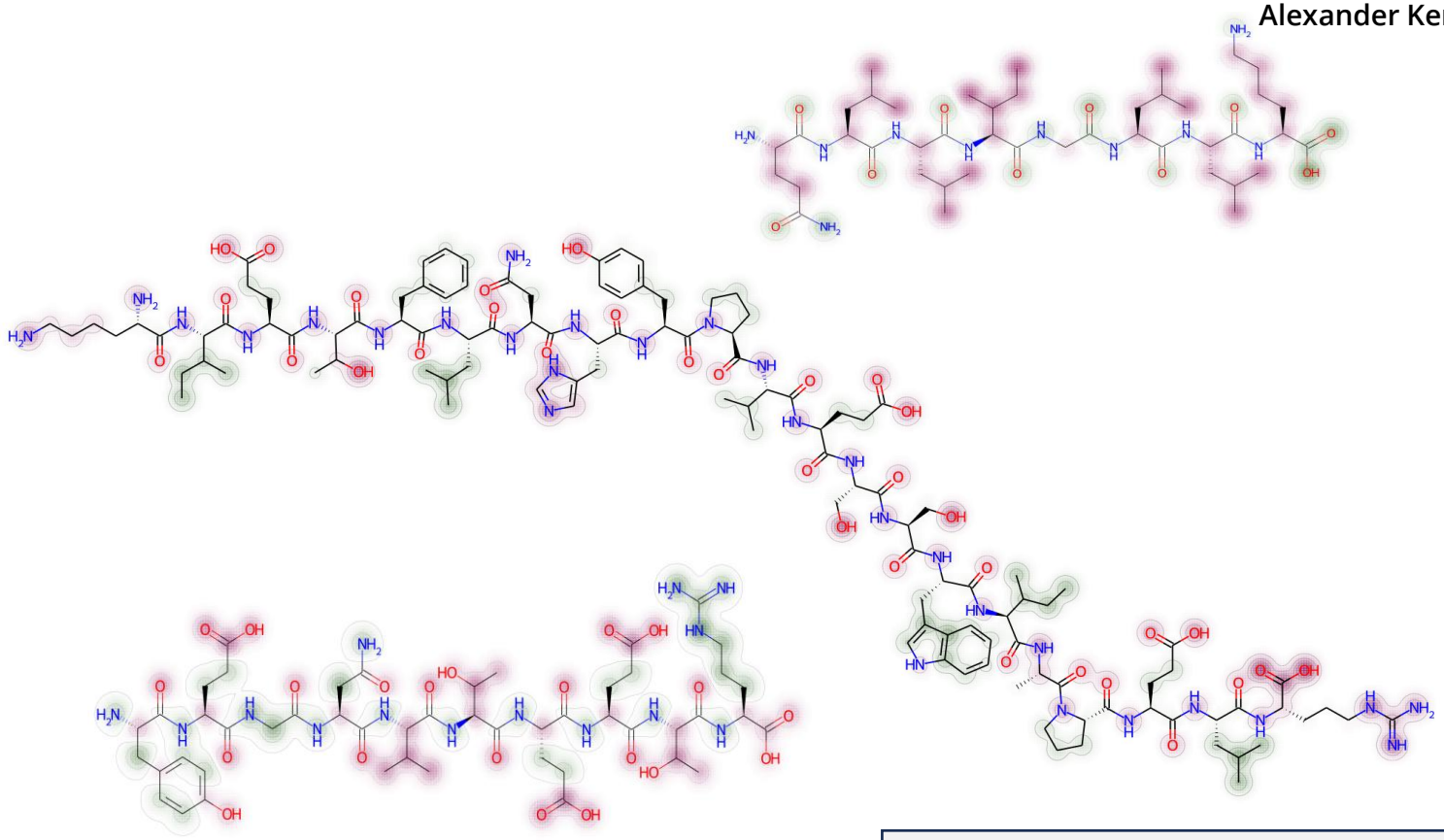
MAE_{DeepLC}: 2.54

MAE (Imp.): 2.28 (-11%)



SWATH Library

N = 11,321



Positive retention time contribution

Negative retention time contribution

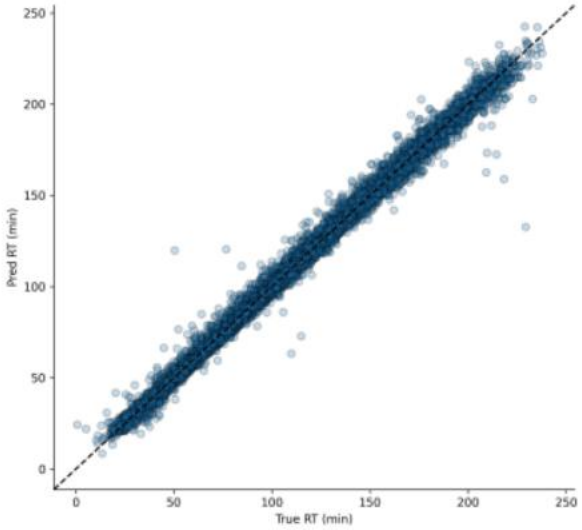
Graph representation slightly improves performance; more importantly structure contribution interpretation



Alexander Kensert

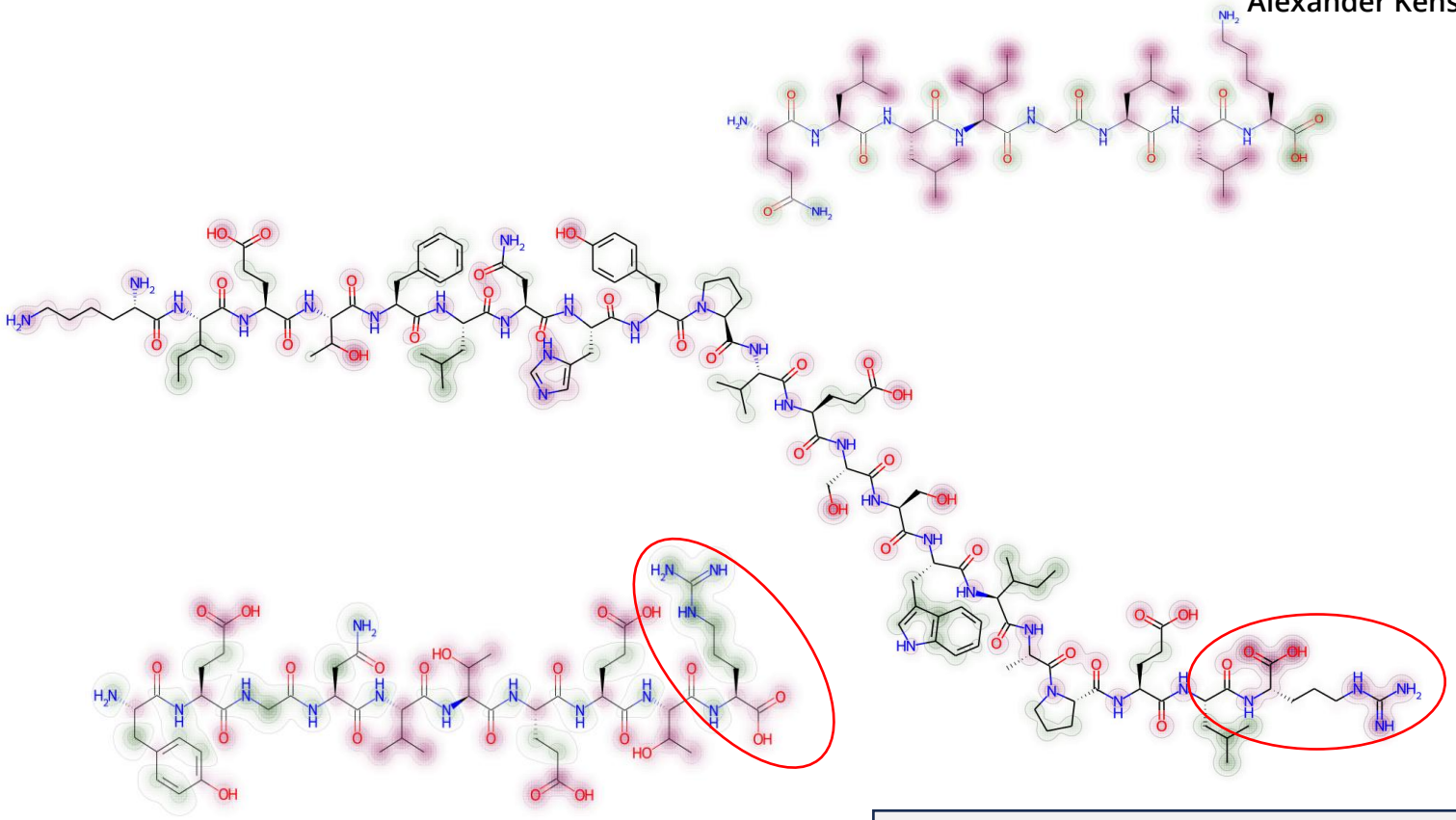
MAE_{DeepLC}: 2.54

MAE (Imp.): 2.28 (-11%)



SWATH Library

N = 11,321



Positive retention time contribution

Negative retention time contribution

Saliency maps to gain fundamental knowledge on the retention mechanism

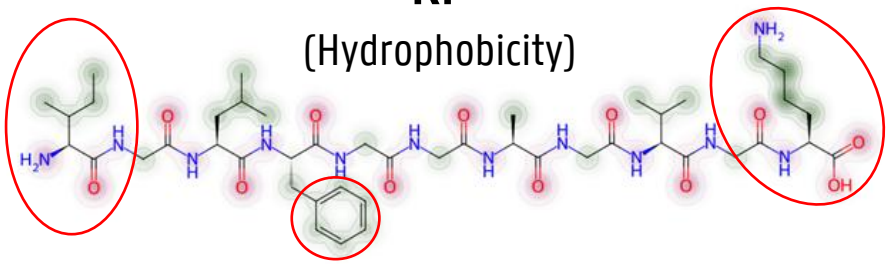
Masaryk University
Faculty of Science



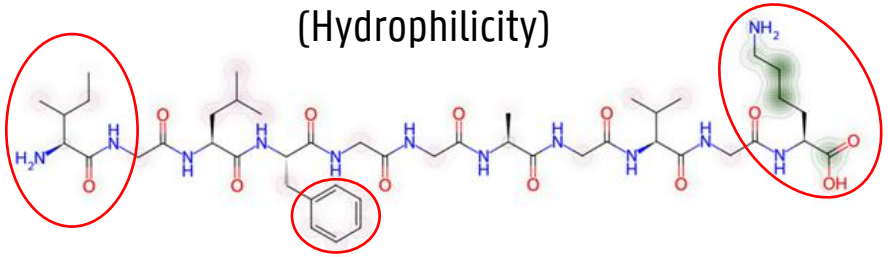


Jiří Urban
Katerina Hruzova

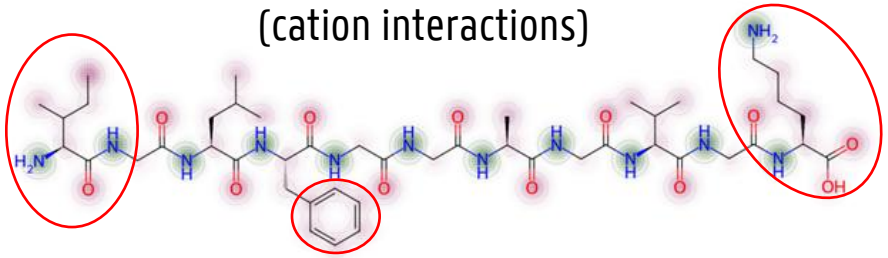
RP
(Hydrophobicity)



HILIC
(Hydrophilicity)



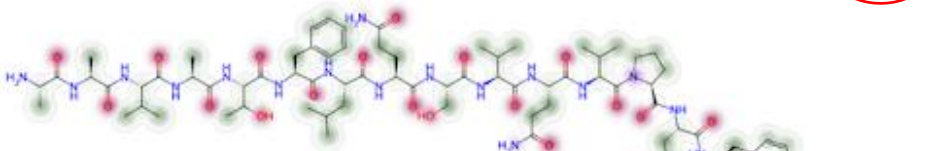
SCX
(cation interactions)



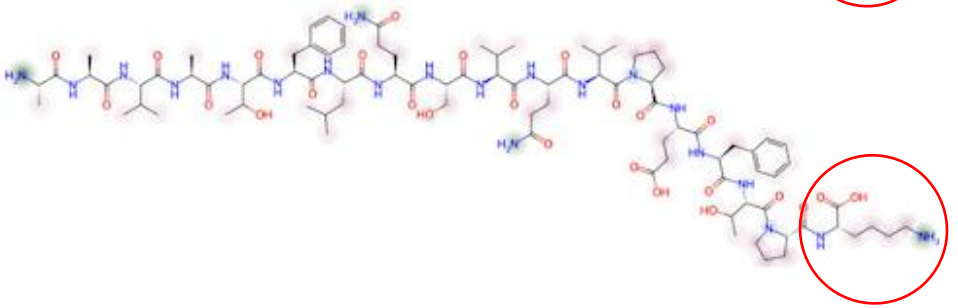
FA
(Formic Acid)



AA
(Acetic Acid)



EG
(Ethylene Glycol)

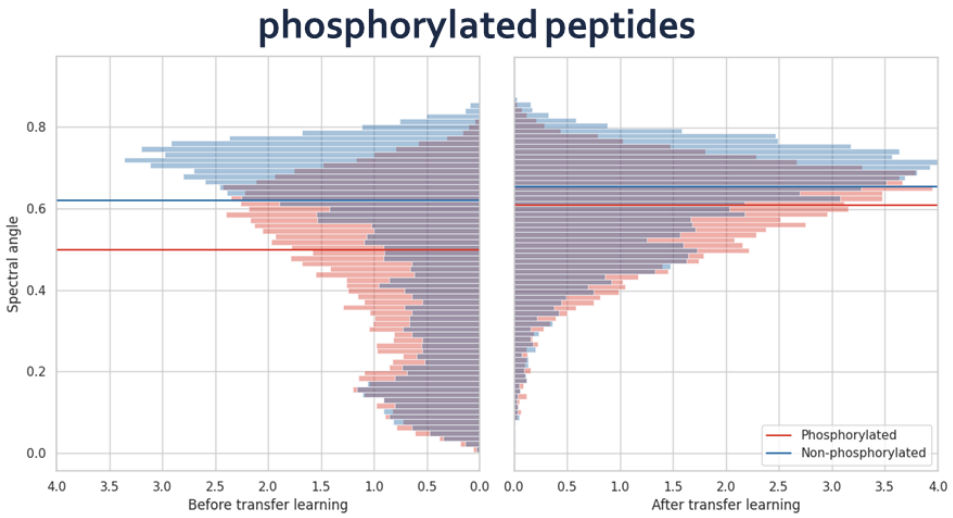


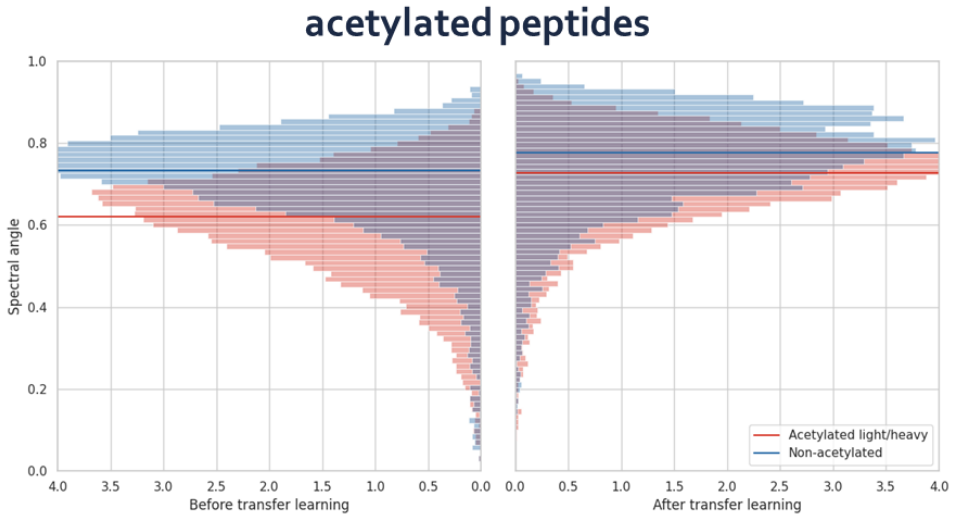
Positive retention time contribution

Negative retention time contribution

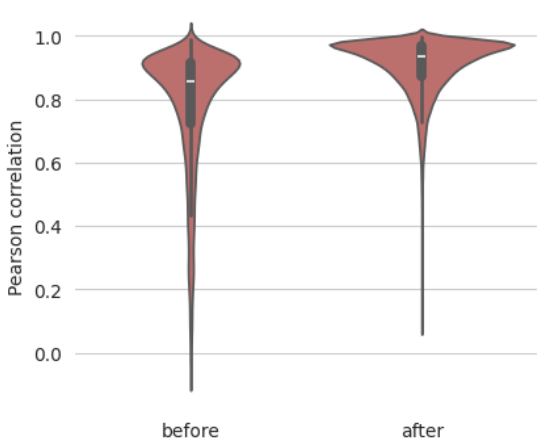
The GNNs also bring transfer learning capabilities, allowing these to adapt even better to diversity

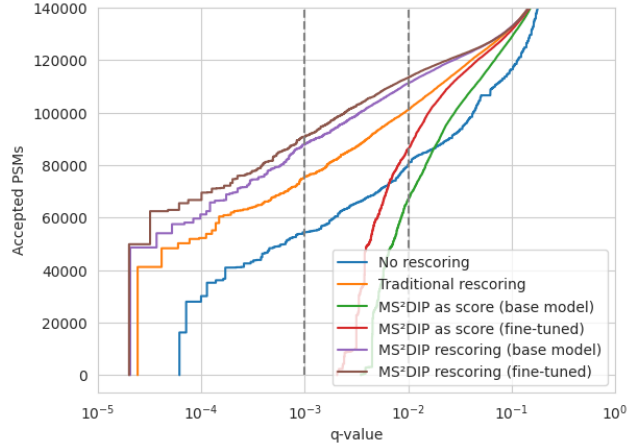






Orbitrap Astral





Gabriels, *in preparation*

So, we built an online, free, open, and easy-to-use tool to provide proteomics metadata: lesSDRF



Ontologies

SDRF guidelines

Local metadata

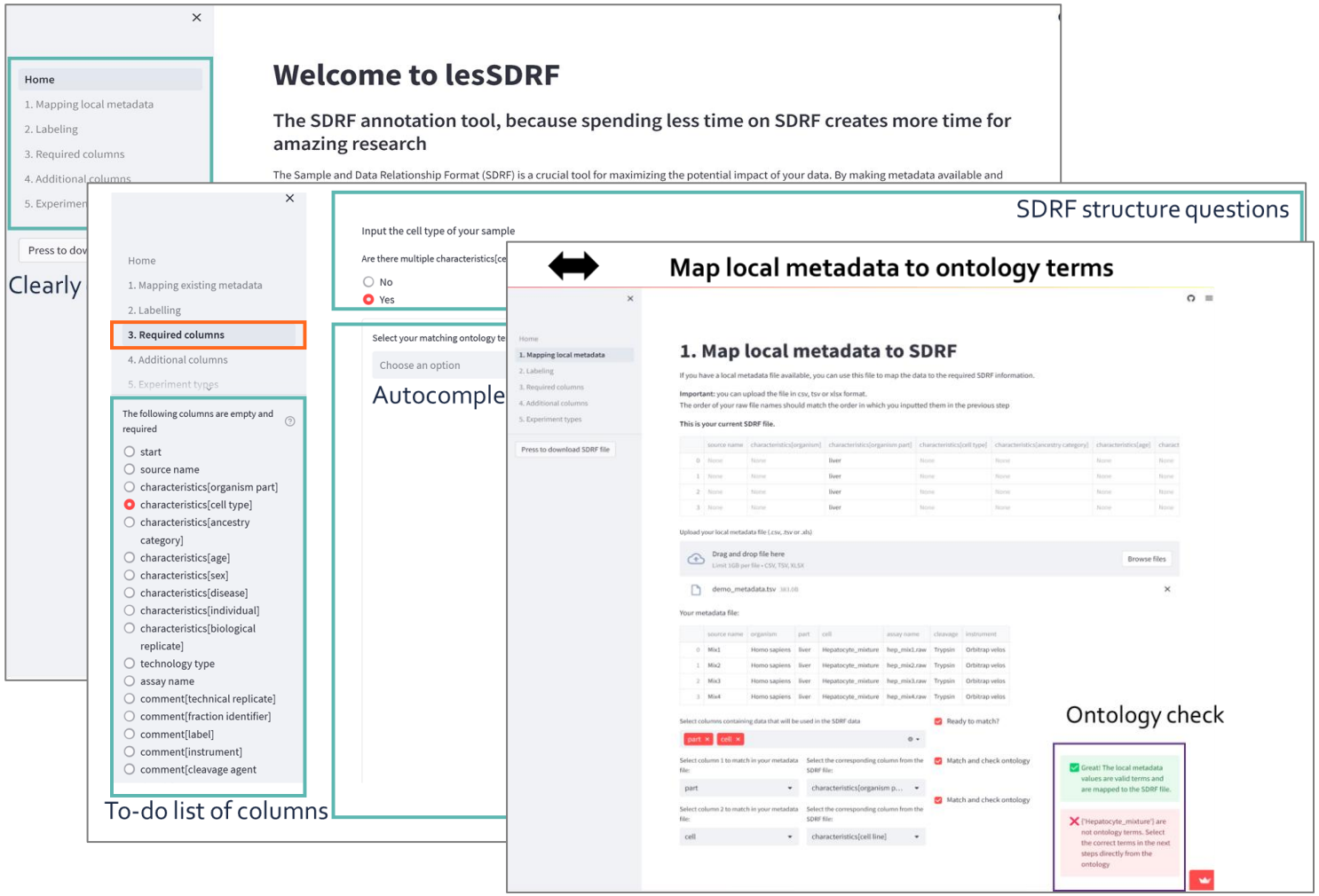
lesSDRF

freely available

open source

no install required

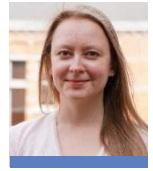
https://lessdrf.streamlit.app/



https://github.com/compomics/lesSDRF

Claeys, Nature Communications, 2023

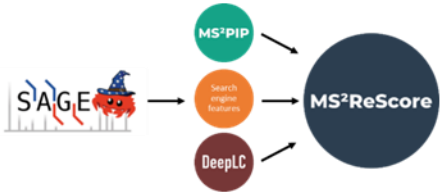
Public human and mouse data reprocessing using our AI tools reveals a hidden and diverse world of protein modifications


PRIDE ProteomeXchange
PRoteomics IDEntifications Database

277 mouse data sets (15%)
appr. **600 million** spectra
14.5 k raw files

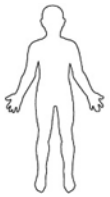


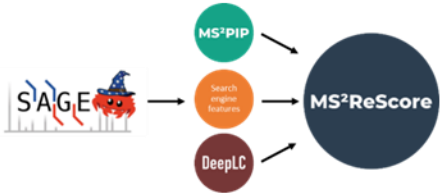


2.9 million modified sites
16 808 proteins
(99% proteome coverage)

PRIDE ProteomeXchange
PRoteomics IDEntifications Database

539 human data sets (5%)
appr. **924 million** spectra
25 k raw files



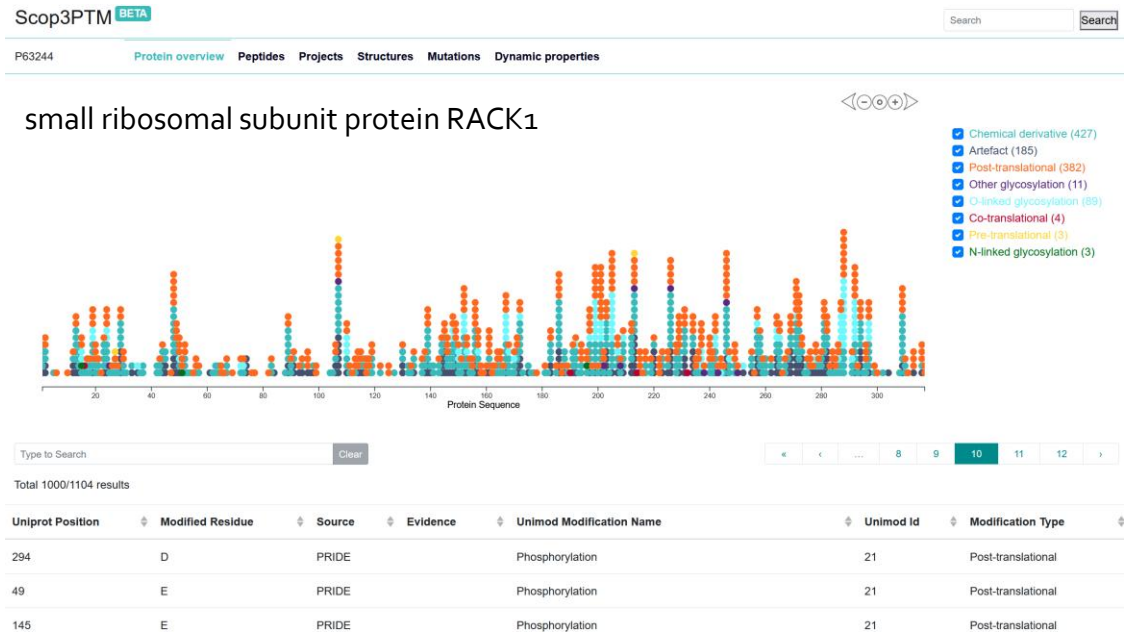


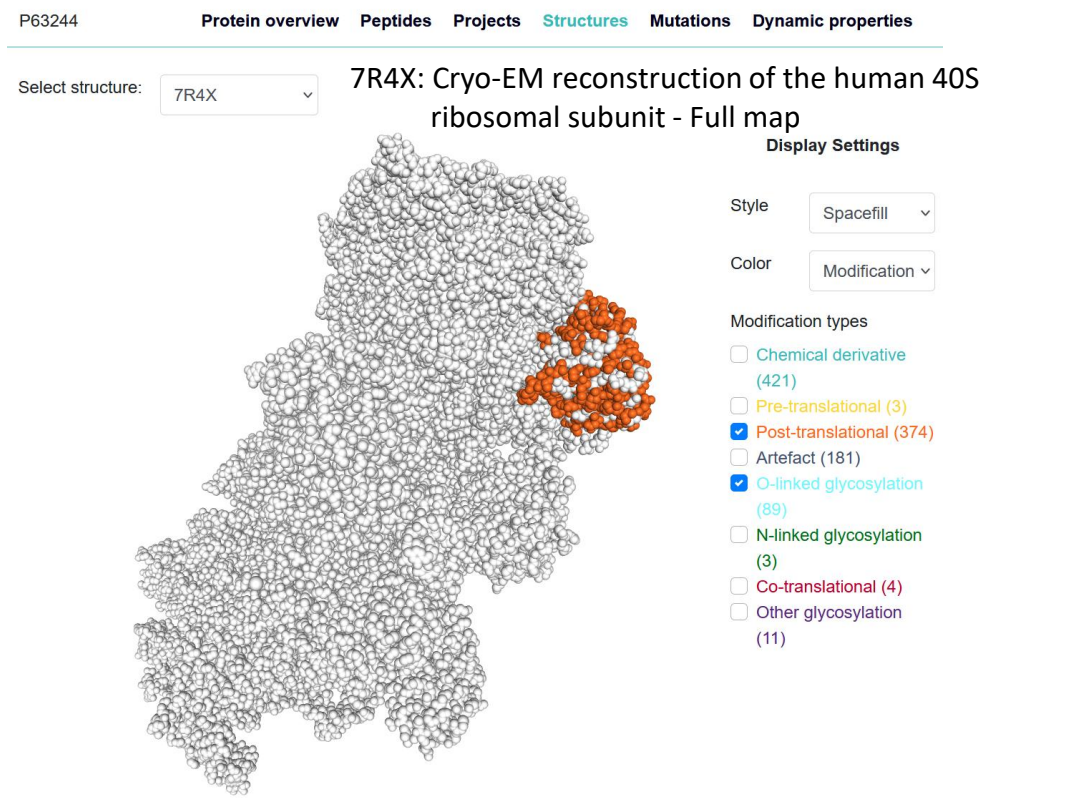
5.0 million modified sites
20 205 proteins
(99% proteome coverage)

PRIDE will be fully reprocessed using our enhanced ML tools and Scop3PTM disseminates the results







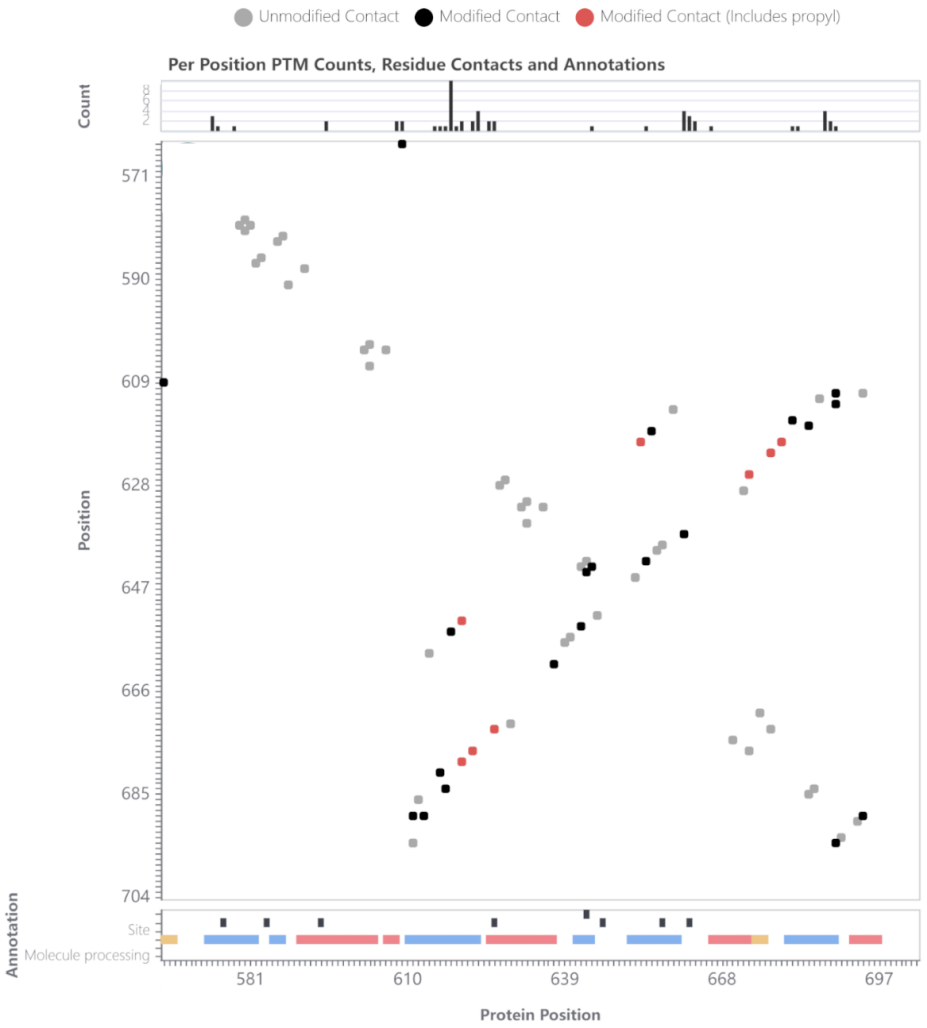


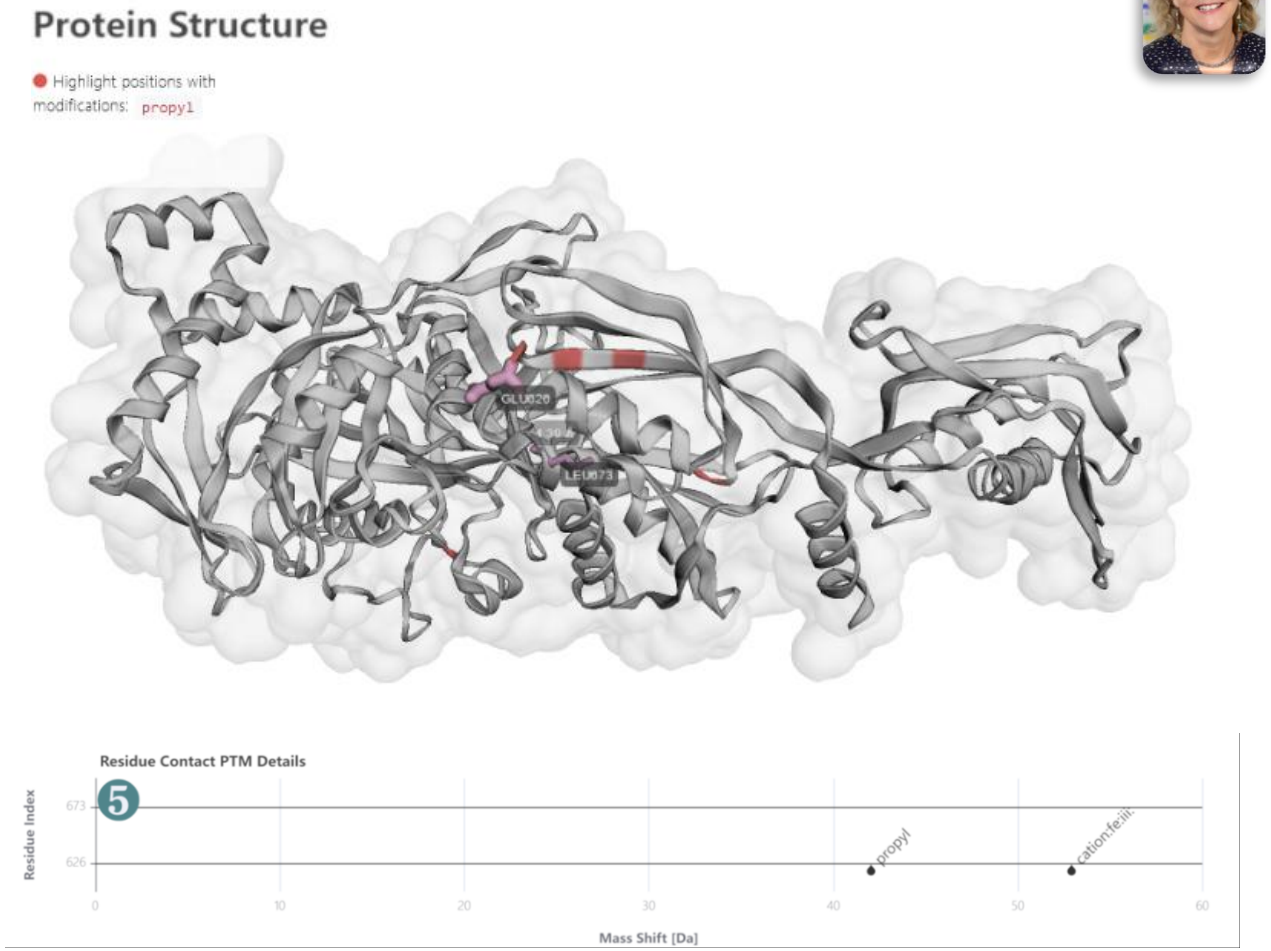
https://iomics.ugent.be/scop3ptm/P63244/summary

Ramasamy, *in preparation*

PTMVision provides really interesting ways to visualise your proteome-wide PTM discovery data



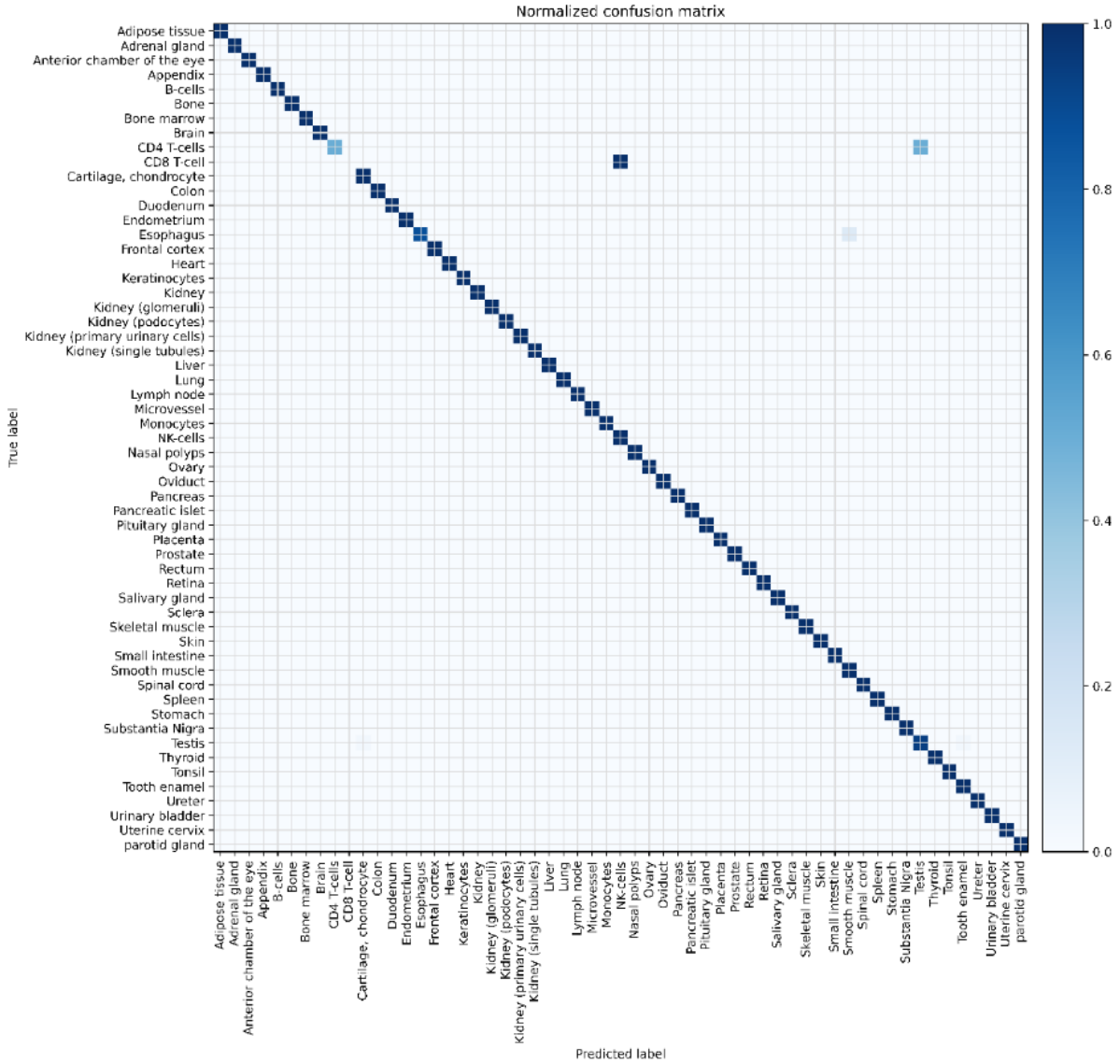




https://ptmvision-tuevis.cs.uni-tuebingen.de/
Hackl, Journal of Proteome Research, 2025

We built a machine learning-based tissue predictor, capable of knowledge extraction and hypothesis generation



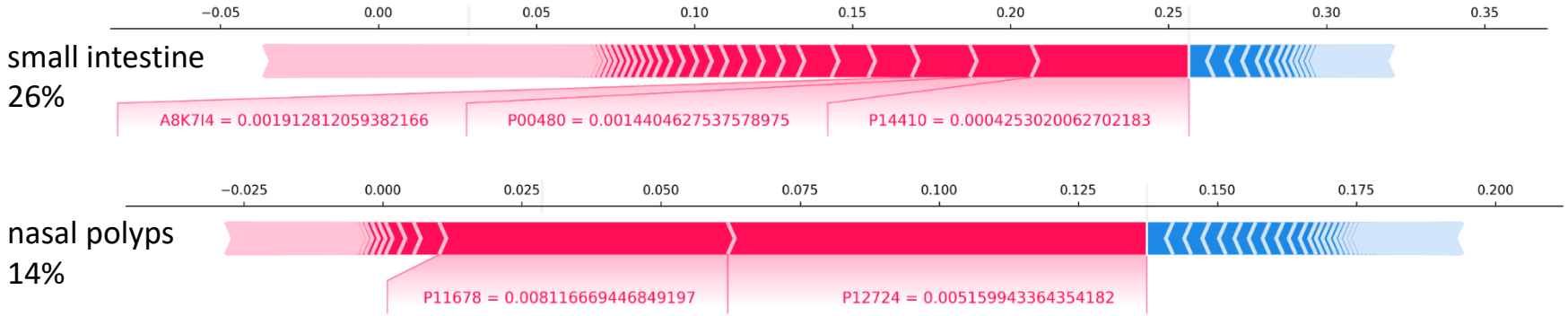


Claeys, J Prot Res, 2022

Strikingly, the predictor functions as a reliable detector of the molecular changes in a diseased tissue



With **celiac disease** proteome (PXD008992) as input



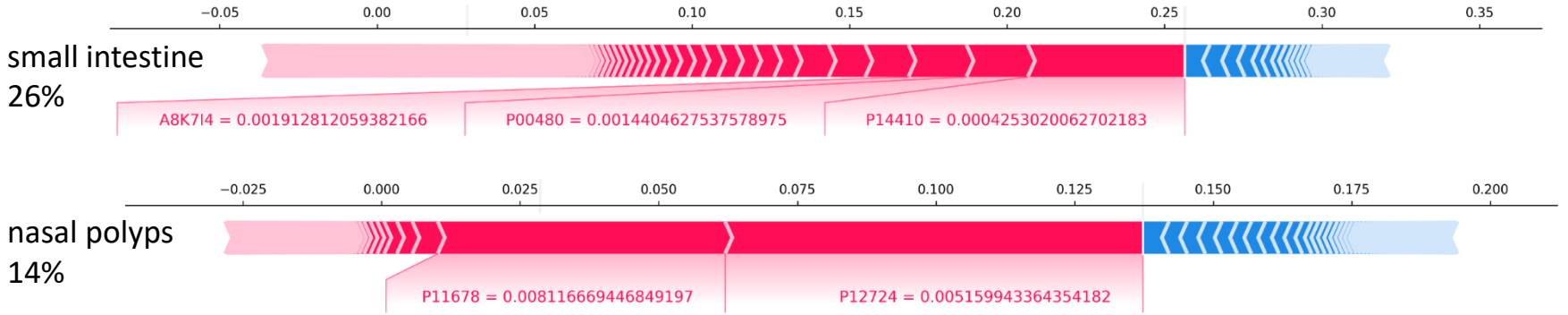
Protein	Protein name	Tissue specificity (HPA, Uniprot)
P14410	Sucrase-isomaltase	Small intestine specific
Q99795	Cell surface A33 antigen	Gastrointestinal epithelium
P12724	Eosinophil cationic protein	Immune related (eosinophiles)
P11678	Eosinophil peroxidase	Immune related (eosinophiles)
Q05315	Galectin-10	Immune related (eosinophiles), **known celiac disease marker**

Claeys, *submitted*, *https://www.biorxiv.org/content/10.1101/2025.06.10.658642v1*

Strikingly, the predictor functions as a reliable detector of the molecular changes in a diseased tissue



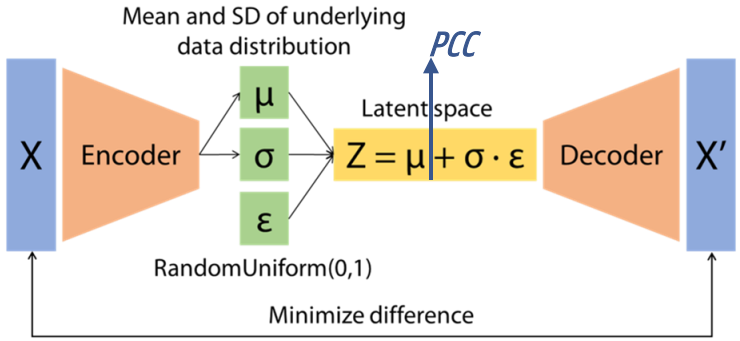
With **celiac disease** proteome (PXD008992) as input

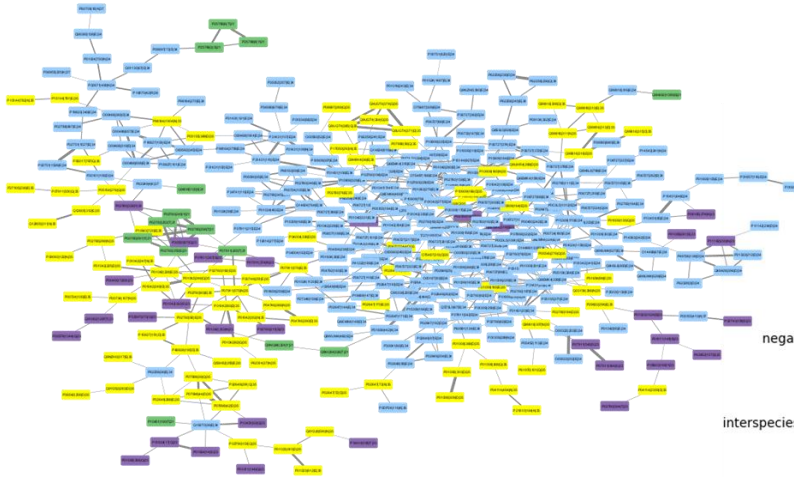


Protein	Protein name	Tissue specificity (HPA, Uniprot)	Original differential expression analysis
P14410	Sucrase-isomaltase	Small intestine specific	Significantly downregulated
Q99795	Cell surface A33 antigen	Gastrointestinal epithelium	Below threshold of downregulation
P12724	Eosinophil cationic protein	Immune related (eosinophiles)	Significantly upregulated
P11678	Eosinophil peroxidase	Immune related (eosinophiles)	Significantly upregulated
Q05315	Galectin-10	Immune related (eosinophiles), **known celiac disease marker**	**Below threshold of upregulation**

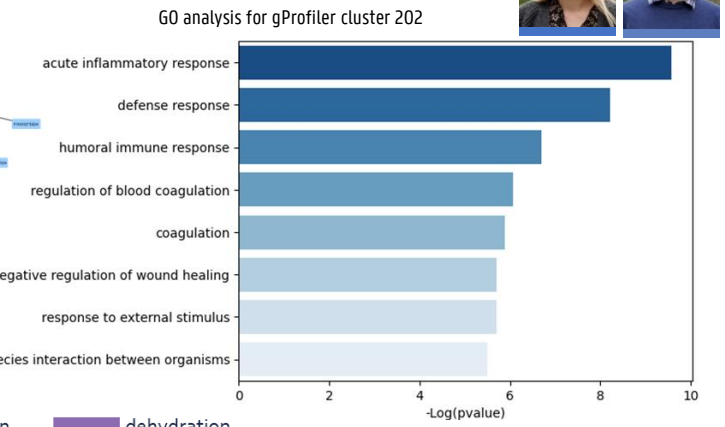
Claeys, *submitted*, *https://www.biorxiv.org/content/10.1101/2025.06.10.658642v1*

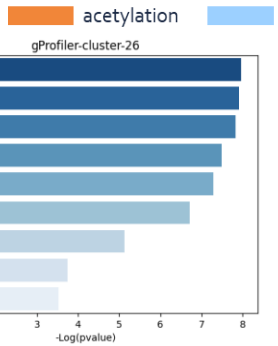
Modification-dependent protein association (MoDPA) networks have been built using a modified FAVA approach

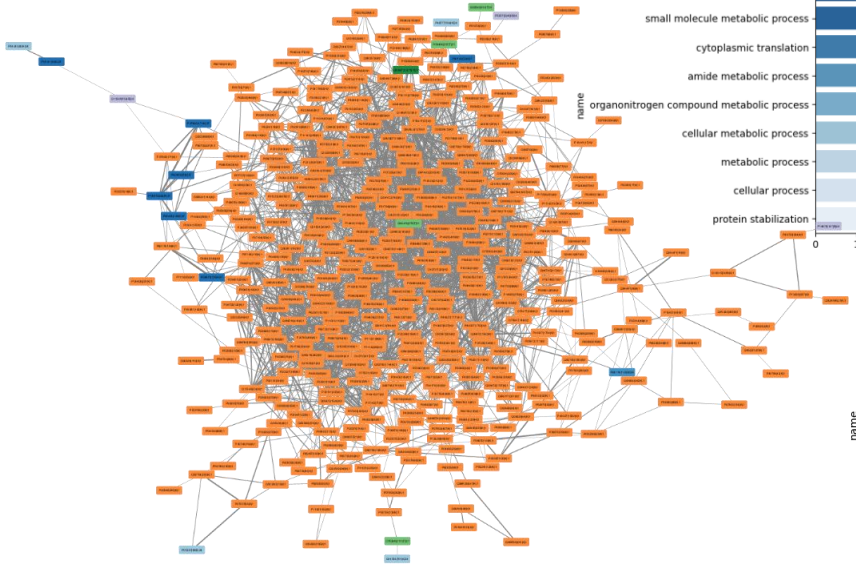


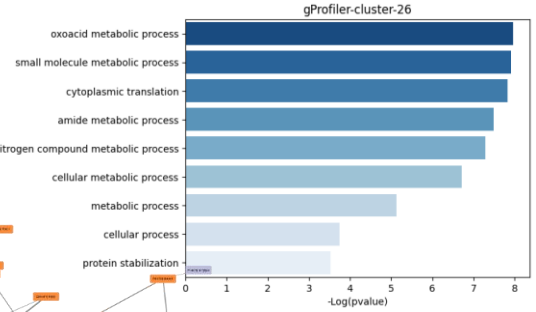


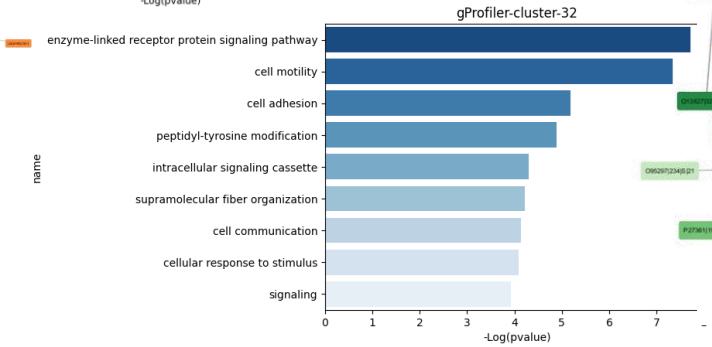


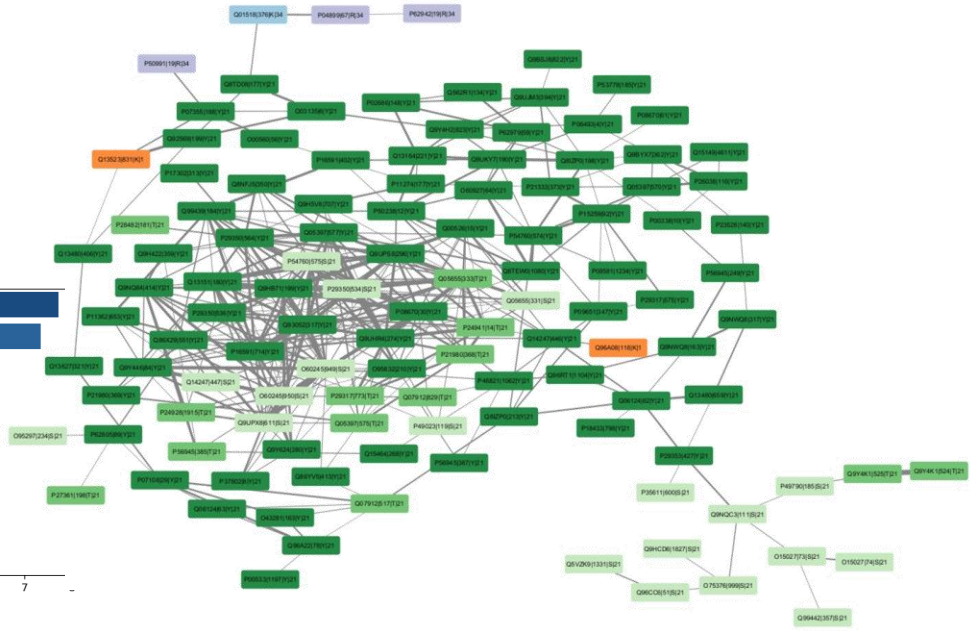












Koutrouli, *Bioinformatics, 2024*

Our online Tabloid Proteome has been updated to disseminate our MoDPA results



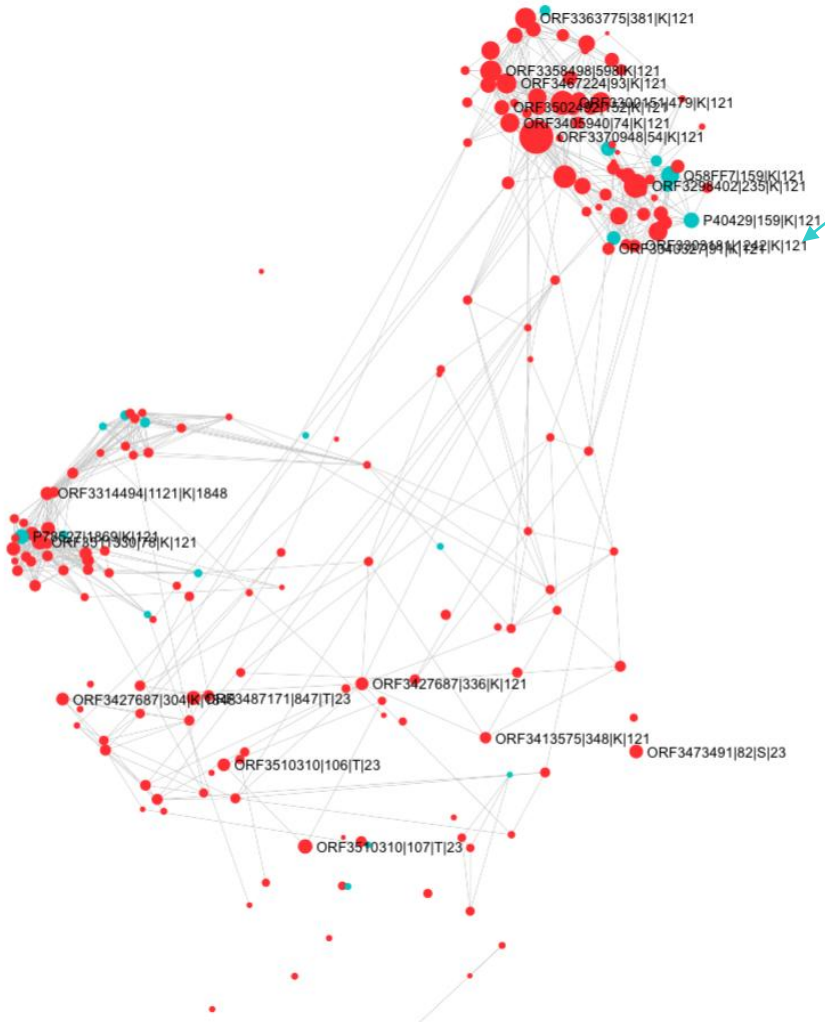




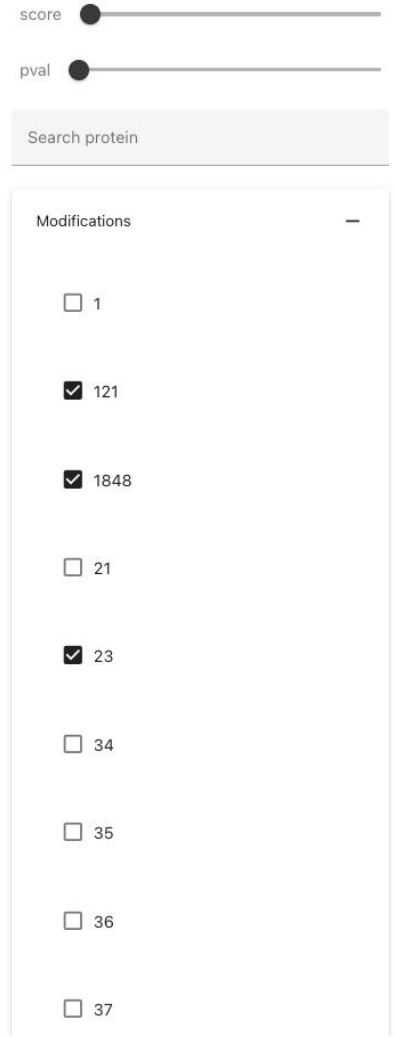
http://www.tabloidproteome.com
Tichshenko and Massignani, *in preparation*

The new TabloidProteome also brings insights into alternative gene products, by incorporating sORFs.





large ribosomal subunit protein uL13
→ *known ubiquitinylation and acetylation on K159*



http://www.tabloidproteome.com
Tichshenko and Massignani, *in preparation*

Conclusion and outlook

Predicted behavior can help to increase sensitivity and specificity rescoring for the identification of (non-tryptic) peptides

Graph neural networks naturally support the structure of peptides and allow for interpretation through saliency maps

Reprocessing allows for deeper interpretation of the proteome, but the right tools are required to visualize and interpret the data

Be ready for the AI wave that will empower (non-)coders to analyze proteomics data



Alexander Kensert



Alireza Nameni



Arthur Declercq



Caroline Jachmann



Enrico Massignani



Harikrishnan Ramadasan



Jasper Zuallaert



Kevin Velghe



Lyudmila Kovalchuke



Natalia Tichshenko



Pathmanaban Ramasamy



Patricia Lefèvre



Pieter Verschaffelt



Ralf Gabriels



Robbe Devreese



Lennart Martens



Sakina Bombaywala



Tine Claeys



Sander Heyndrickx



Tanja Holstein



Tim Van Den Bossche



Sam van Puyenbroeck



Toon Callens



Xuxa Malliet

VIB

GHENT UNIVERSITY

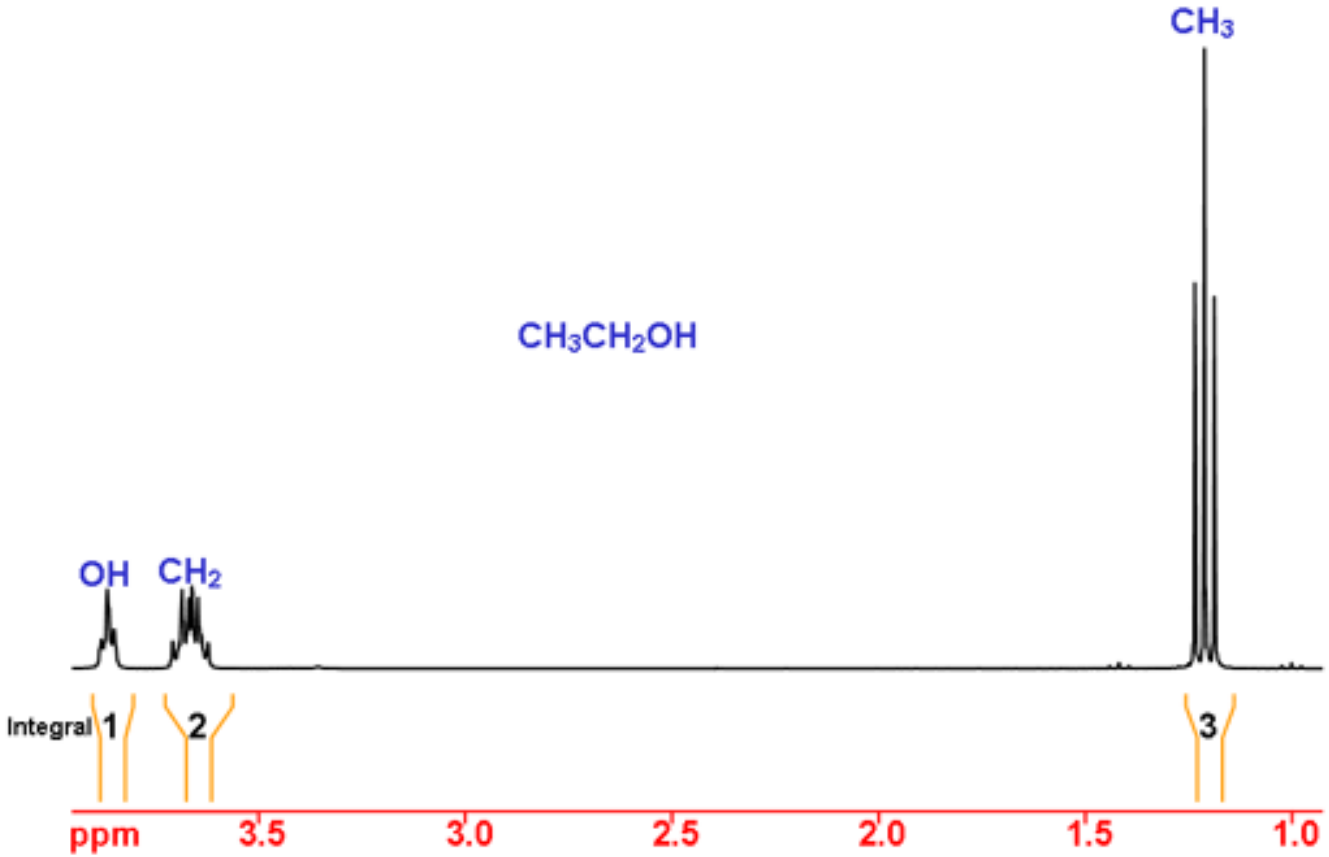
Comp omics

LSMBO

cnrs

Université de Strasbourg

Nuclear magnetic resonance (NMR) measures signals from excited atomic nuclei



In a strong magnetic field, radiofrequency pulses excite nuclei.

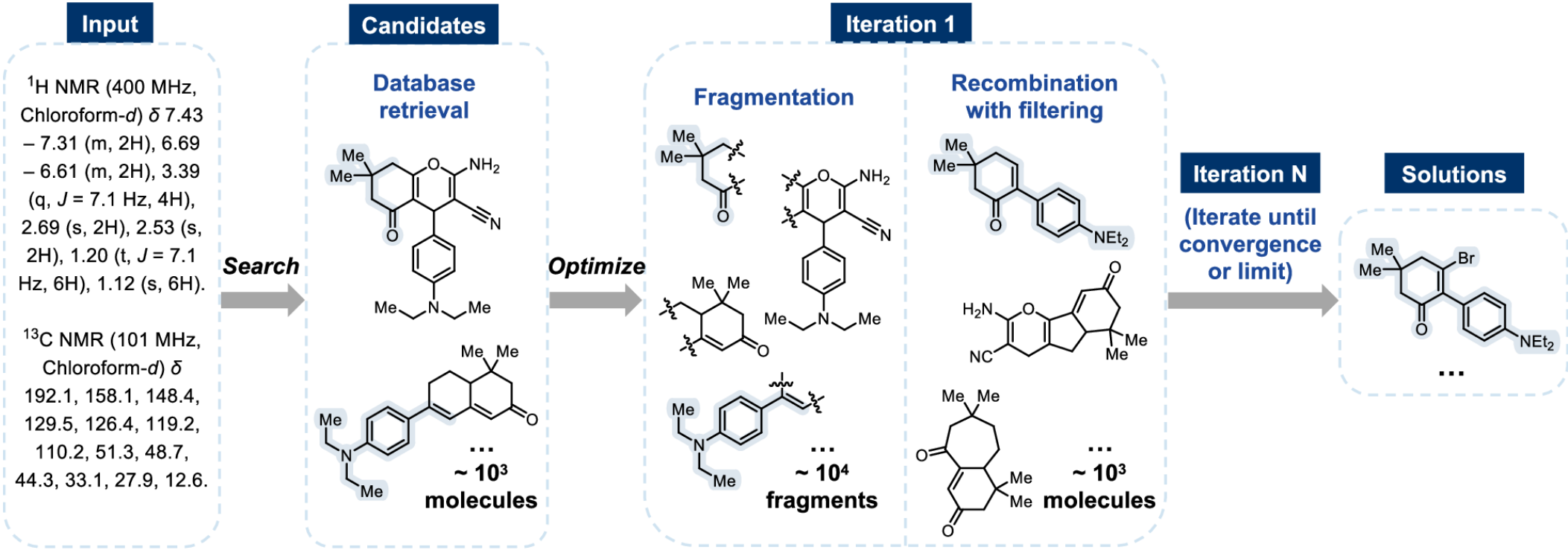
Position: chemical surroundings.
Area in ^{1}H NMR: relative number of hydrogens.

Goal: find a molecular structure that explains the spectrum.

Measured ^{1}H NMR spectrum of ethanol in $CDCl_3$. Hebrew University NMR laboratory, Fig. 2.

NMR-solver proposes a molecular structure based on an input spectrum with a transformer architecture

Problem: which molecule explains the measured ^{1}H and ^{13}C chemical shifts?



1 Find starting candidates
Search a database of predicted shifts.

2 Try related molecules
Recombine molecular fragments.

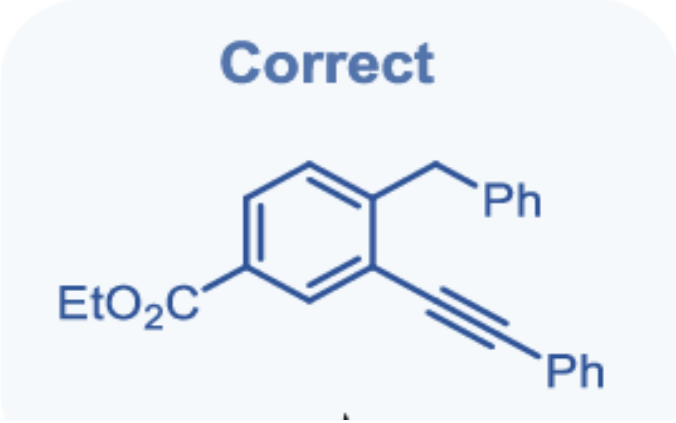
3 Compare and repeat
A neural network (transformer) predicts the shifts.

Jin et al. (2026), Fig. 2. CC BY 4.0

NMR-solver proposes a molecular structure based on an input spectrum

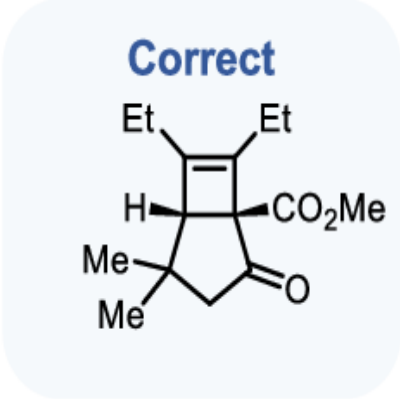
Results include laboratory assignments and a documented wrong prediction.

A laboratory product

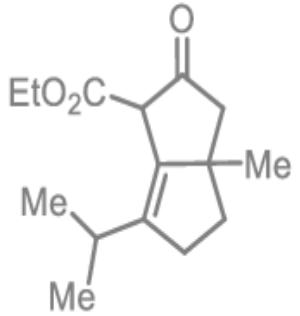


Additional 2D NMR and high-resolution mass spectrometry supported this assignment.

A strained molecule



Actual product



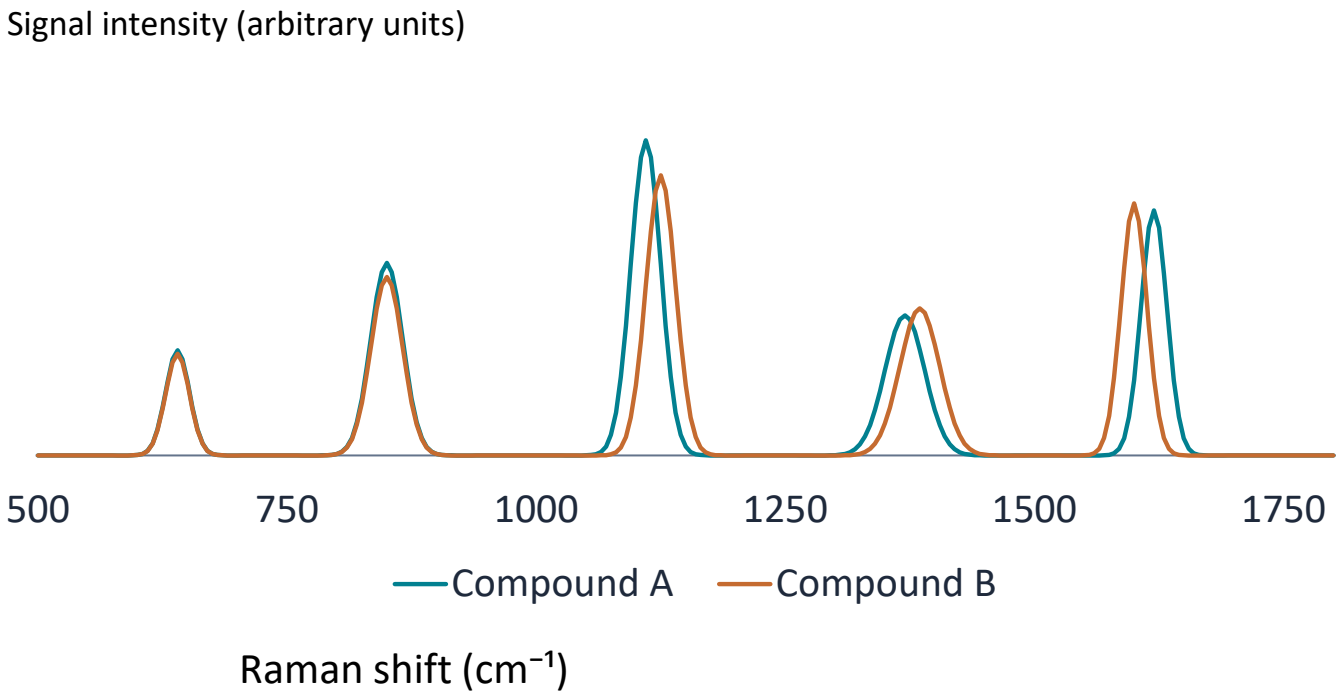
Model prediction

Inaccurate predicted shifts made the wrong structure look more convincing.

Jin et al. (2026), extracts from Fig. 4a and 4f. CC BY 4.0

Raman fingerprints molecular vibrations based scattered laser light changes in energy

Similar compounds can share many peaks



What produces the signal?

Laser light exchanges energy with molecular vibrations.

What do the peaks tell us?

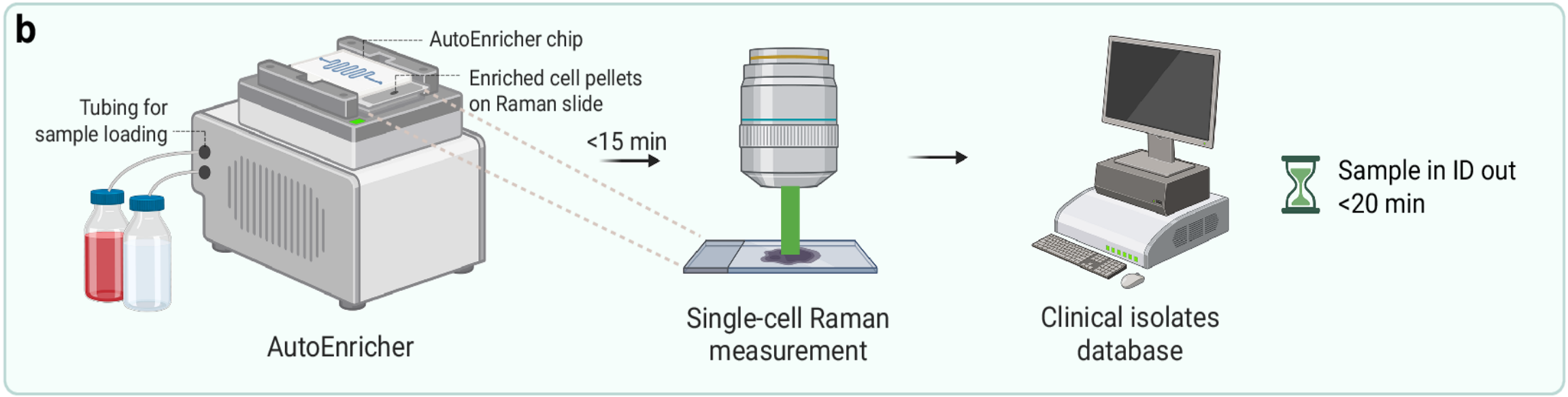
Peak positions reflect vibrations. The overall pattern helps distinguish compounds.

Goal here: identify microorganisms from single-cell Raman fingerprints.

Concept: Renishaw, What is Raman scattering? Schematic spectra of hypothetical compounds.

Raman can be used for identification of microorganisms

Question: which microorganism is present in the sample?



1 Enrich the sample
Collect microbial cells.

2 Learn spectral patterns
A CNN learns from known species.

3 Combine cell predictions
Several cells contribute to each result.

Concept: Renishaw, What is Raman scattering? Schematic spectra of hypothetical compounds.

Certain species have bad performance; mostly a limitation of the data and ability to discriminate

95.1%

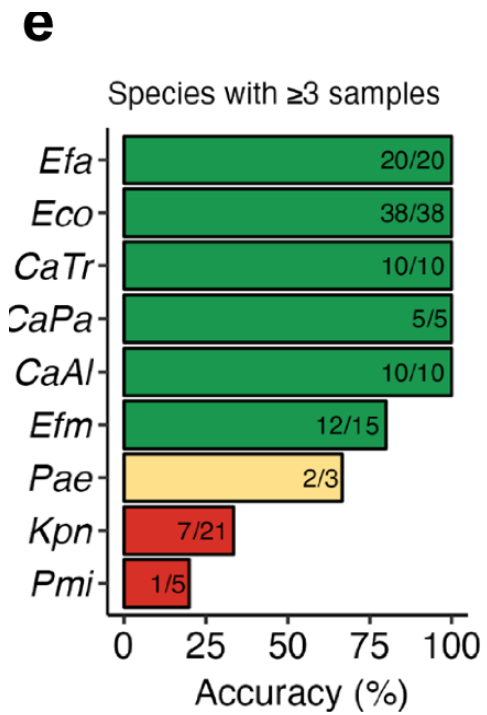
Accuracy across 342 clinical isolates

Validation keeps each isolate's spectra together.

82.5%

99 of 120 selected positive samples agreed with culture and MALDI-TOF identification.

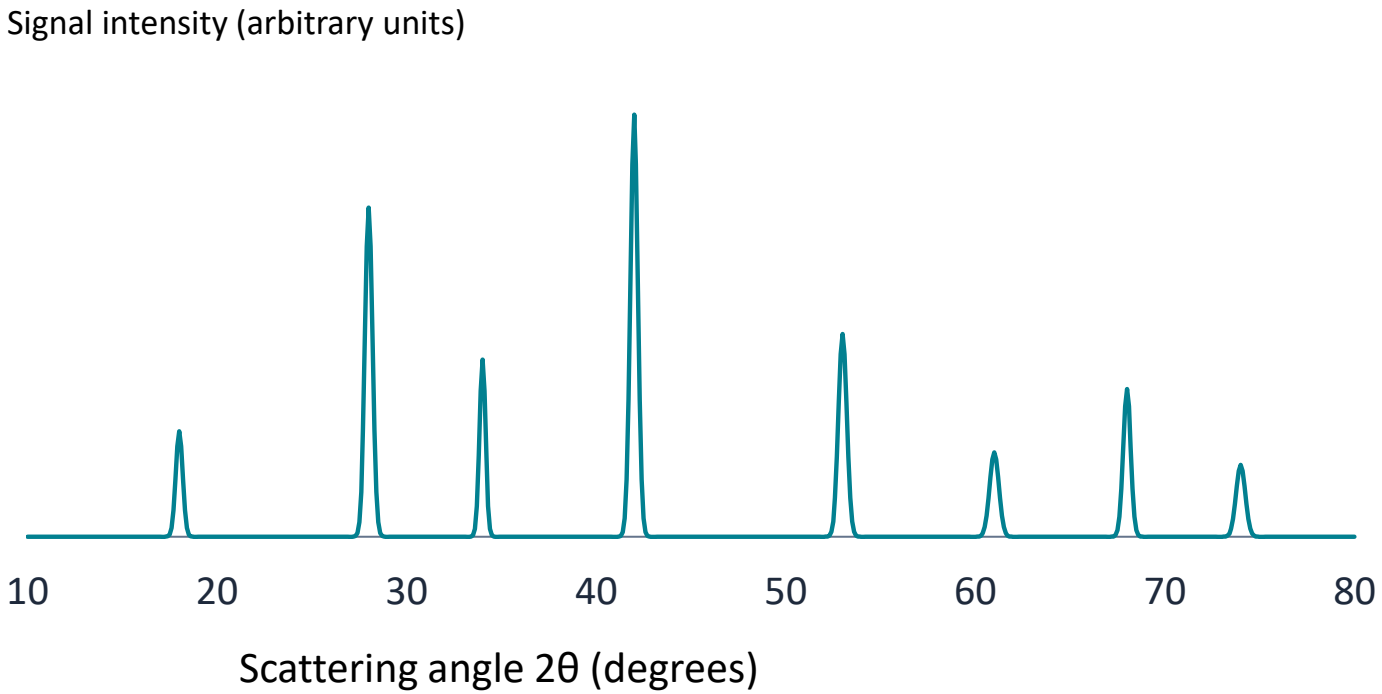
Some species perform poorly



Li et al., Nature Communications 17, 283 (2026), Fig. 8e, cropped. CC BY 4.0

Powder diffraction (X-rays) creates spectra based on angles it was diffracted

Intensity at different scattering angles



Atoms scatter X-rays. Waves reinforce each other at certain angles.

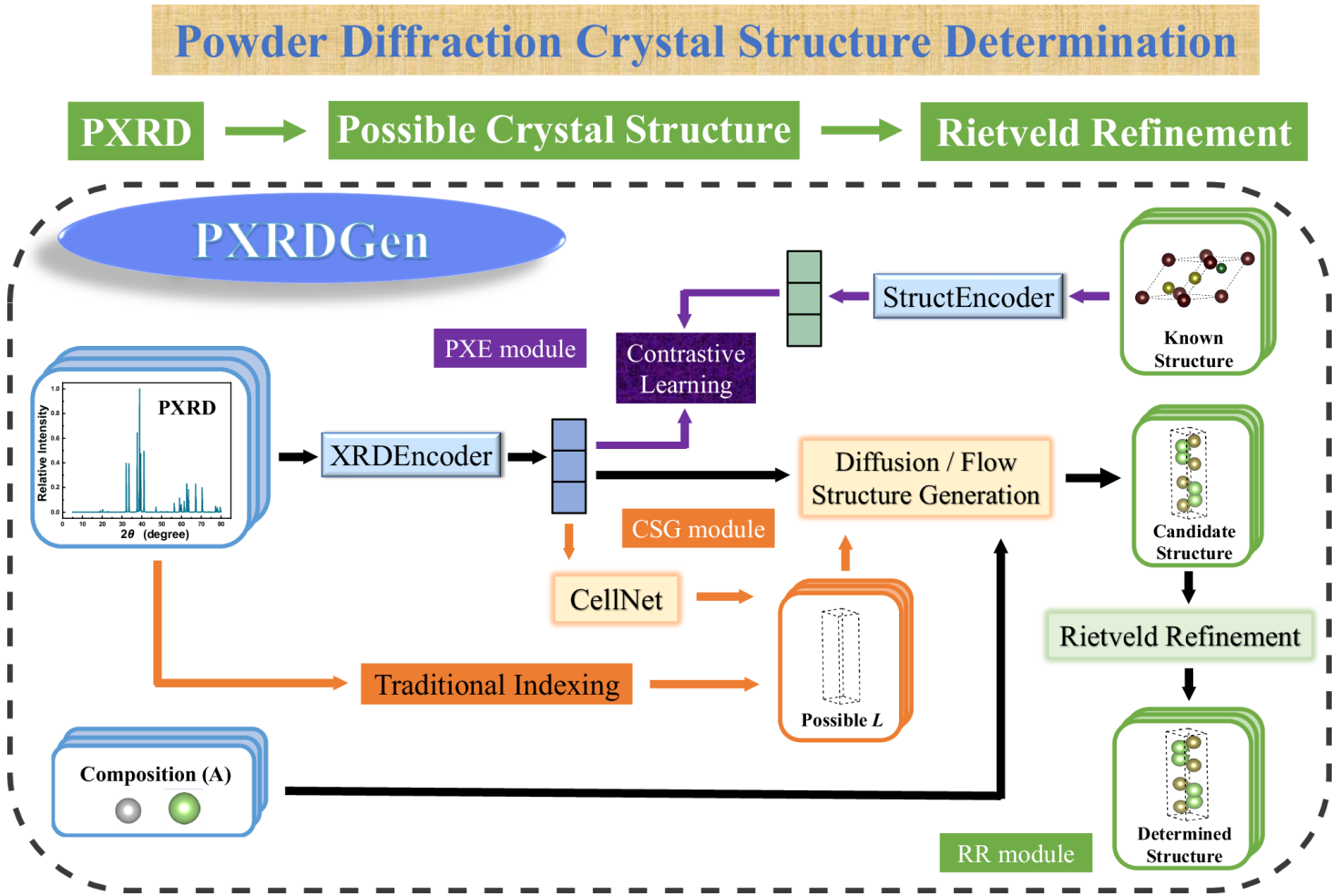
Positions reflect repeating spacings. Intensities depend on the atoms and their positions.

Goal here: use the pattern and formula to infer an atomic arrangement.

Concept: IUCr teaching pamphlet, An introduction to X-ray diffraction. Schematic pattern.

PXRDGen resolves atomic arrangements from a powder to get a structure

Problem: overlapping diffraction peaks make crystal structures difficult to solve.



Read the pattern

A neural network extracts information from the diffraction peaks.

Propose atomic arrangements

A generative model uses the pattern and chemical formula.

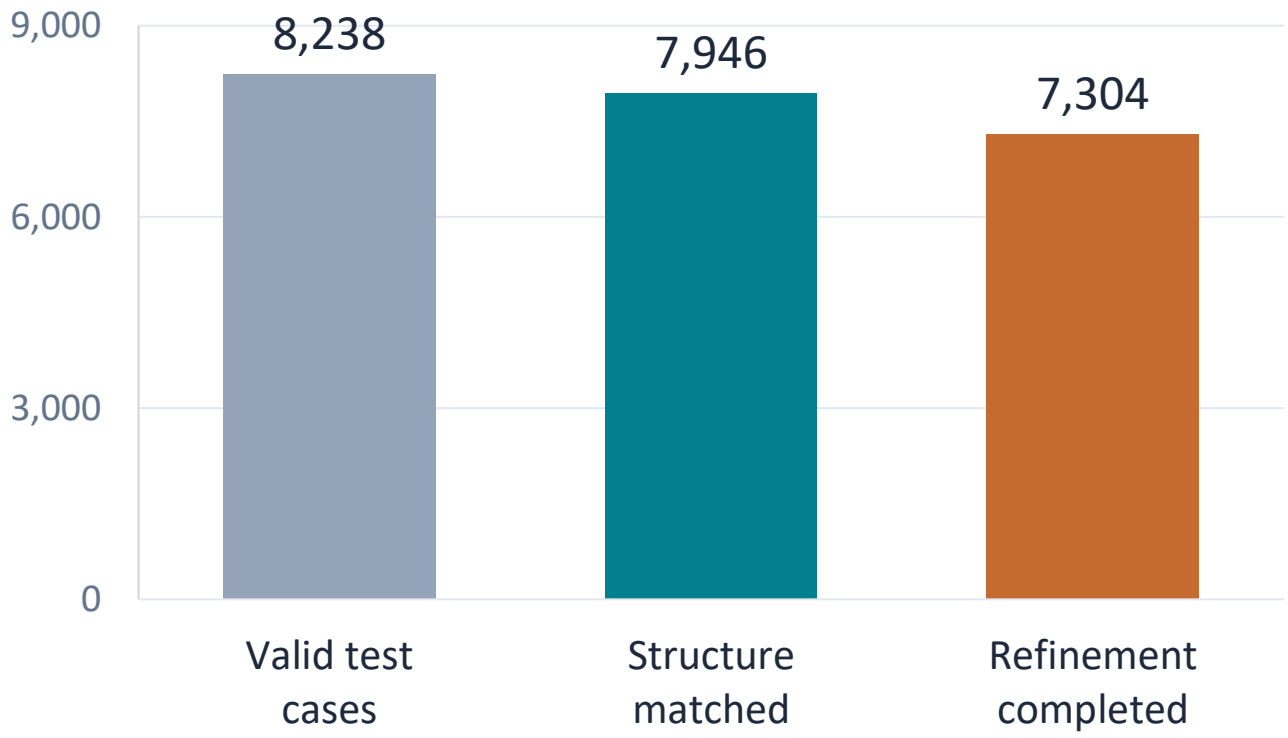
Refine against the data

Rietveld refinement adjusts the structure to improve the fit.

Li et al. (2025), Fig. 1 reproduced unchanged. CC BY-NC-ND 4.0

PXRDGen resolves atomic arrangements from a powder to get a structure

MP-20 benchmark with simulated diffraction patterns



A close diffraction fit still needs chemical assessment.

Benchmark conditions

Known unit-cell parameters.
20 candidate structures per case.

Scope

The training benchmark contains inorganic crystals with at most 20 atoms per primitive cell.

Complex oxides and organic crystals need further development.

Li et al. (2025), results underlying Fig. 5. Counts redrawn from the text.